

Graph Machine Learning for Asset Pricing: Traversing the Supply Chain*

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Abstract

We develop a graph neural network (GNN) framework for asset pricing with supply chains, nonlinearly aggregating firm-level information across d -hop neighborhoods. We find that depth $d = 3$ empirically balances bias reduction from higher-order relations against variance from added complexity; beyond $d = 3$, oversmoothing results in homogeneous embeddings and collapses predictive R^2 from 15.3% to near zero for $d \geq 8$. We leverage rich benchmarks to residualize the supply chain GNN-managed portfolio, which yields an out-of-sample Sharpe ratio of 1.07. Sensitivity to missing links concentrates in high-centrality rather than large-cap firms, consistent with the GNN learning from graph topology rather than firm size.

Keywords: Supply chain, cross-section of expected returns, machine learning, graph learning, interpretability of machine learning.

JEL classification: C33, C38, C45, C52, C55, G12

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1 Introduction

A central challenge in empirical asset pricing is how to incorporate the relational structure of the economy—the customer–supplier linkages that connect firms through supply chain networks—into conditional mean return estimation. Standard panel-data models treat firms as independent cross-sectional units, discarding the graph topology of the supply chain entirely. Yet input-output shocks propagate through the supply chain network, generating forecastable return co-movement that own-firm models cannot detect (Cohen and Frazzini, 2008; Ding et al., 2021).¹ Exploiting this channel for asset pricing poses a fundamental methodological challenge: the supply chain network is a graph, and standard panel-data models are ill-equipped to represent, let alone leverage, its topology. We introduce the first graph neural network (GNN) framework for systematic asset pricing with supply chains, providing a principled method for integrating supply chain graph structure with the rich panel of firm characteristics studied in the asset pricing literature. Using this framework, we identify the supply chain network as a structural source of cross-asset return predictability—the forecastable co-movement in returns across firms recently documented empirically (Kelly et al., 2025, 2023) but whose economic mechanism is not yet fully understood.

Our central methodological contribution is a novel message-passing architecture that estimates a return kernel by integrating firm characteristics-managed portfolio returns as the model traverses the supply chain graph (SCG). Unlike prior work conditioning solely on direct customer–supplier links (Cohen and Frazzini, 2008; Ding et al., 2021), our framework captures higher-order transmission effects through nonlinear aggregation across the full graph topology. These effects are, by construction, unspanned by own-firm or pairwise models. The empirical design addresses Cochrane’s (2011) challenge of finding “*which characteristics really provide independent information about average returns ... (vs) subsumed by others*”. We conduct systematic comparisons against a comprehensive set of benchmark models that are also conditioned on the rich characteristic set of Freyberger et al. (2020). We assess the economic significance through actively managed long-short portfolios, and test whether the supply chain GNN captures priced risks that are incremental to those already spanned by existing factor models.

Methodologically, supply chain data introduces multiple levels of complexity. Supply chain linkages are often missing from observed data, and their missingness often results from strategic decisions made by firms.² Moreover, supply chains evolve as firms adjust their partnerships. Such

¹Supply chain linkages introduce systemic risks—input shortages, intermediate-goods price surges, and exposure to geopolitical shocks such as the energy crisis triggered by the war in Ukraine (International Energy Agency, 2024; Capponi et al., 2024; Birge et al., 2023)—that have been further amplified by the disruptions experienced during the pandemic (Wall Street Journal, 2024). We refer to Babich and Birge (2021) for a comprehensive review of how operational resilience, risk management, and financial stability are interwoven in modern supply networks.

²Firms’ disclosure behavior of supply chain data is an active field of research in finance, economics and management science. In the supply chain transparency literature, for instance, whenever the disclosures may build brand trust among consumers, firms may choose to disclose supply chain linkages (Mohan et al., 2020). See Sodhi and Tang (2019) for a detailed review of current discussion on transparency. From the financial economics perspective, firms may choose supply chain linkages for non-production reasons. For example, firms may strategically form supply chain relationships for tax avoidance (Cen et al., 2017), accompanied by reduced disclosures. As is common in similar research, we rely on supply chain data from government-imposed disclosures that are subject to firms’ strategic

complications are also often not captured by data, or captured with lags and censoring. Existing asset pricing models lack the ability to incorporate these structural nuances and changing relationships. Finally, one also needs to take into account the rich literature of return anomalies (Freyberger et al., 2020; Hou et al., 2020) when modeling supply chain interconnectedness, making the problem very high-dimensional. Thus, we need to clear a primary hurdle of using supply chain data: how to effectively leverage noisy, dynamic, and high-dimensional supply chain data to estimate a conditional mean model for the returns on the firm level.

Our ML-based framework can utilize supply chain data for asset pricing, tackling the challenges articulated above. We view supply chains with firm characteristics as graphs with covariates. Each firm is a node and each linkage is an edge³, and for each month-firm, the node is associated with a return and a vector of firm characteristics.⁴ Given the inherent incompleteness of supply-chain data, our asset-pricing model relies on signals from observed linkages and is naturally interpreted as conditioning on investors’ observable information. Because our machine learning procedure is non-parametrically adaptive by nonlinearly weighing signals from local neighborhoods of the supply chain graphs, the evolving supply-chain network *helps* the model learn how edge additions and removals affect the GNN’s return estimates.

Empirically, a moderately complex GNN with $d = 3$ outperforms linear supply-chain benchmarks in conditional mean estimation.⁵ Against Neighboring Characteristics (NC) Ridge and LASSO models, which incorporate supply chain information but restrict aggregation to nearest-neighbor firms and forgo the GNN’s nonlinear flexibility, the GNN reduces out-of-sample mean absolute error by 12% (see Table 1 for details). The GNN also dominates a Leontief-inverse regression, which aggregates powers of the adjacency matrix linearly to approximate economic interdependence across the network. This suggests that the nonlinear, node-adaptive GNN is capturing heterogeneous propagation patterns, such as differences in how shocks attenuate, amplify, or redirect across supply chain positions, that fixed polynomial weighting cannot represent.

We find that the GNN model reveals strong out-of-sample decile return separation. The long-minus-short zero-cost portfolio achieves a higher Sharpe ratio by capturing higher-order supply chain relationships in a *nonlinear* way, delivering 16–21% additional out-of-sample Sharpe ratio relative to Ridge and LASSO benchmarks that also condition on the SCG. Our GNN portfolio is constructed by ranking firms according to GNN-implied conditional mean returns, and significantly enhances the maximum Sharpe ratio when traded jointly with factors summarizing firm characteristics. Hence, supply chain signals captured by the GNN make independent contributions to explaining the cross-section of average returns. In a time-series regression controlling

disclosure choices.

³The full definition of SCG’s nodes and edges is given in §2.2. We use undirected, unweighted graphs throughout to focus on the network topology rather than edge magnitudes; this simplification is sufficient to demonstrate the value of graph-structured information for return prediction. Extensions to more complex graphs such as hypergraphs, or sales-weighted graphs are interesting areas for future research.

⁴Modern GNNs are designed for such graphs, and have empirically shown great performance on much larger datasets such as the social network of Reddit (Hamilton et al., 2017) and protein molecules (Shi et al., 2021).

⁵The parameter d is interpretable as the radius of the local supply-chain subgraph the model scans; see §2.4.

for Fama–French factors, the GNN-managed portfolio is statistically significant for 92% of the 25 double-sorted test portfolios from Kenneth R. French’s website. Moreover, the GNN-managed portfolio has low correlations with known risk factors (see Table A.1). These results collectively identify the supply chain network as the structural source of the cross-asset return predictability documented by Kelly et al. (2025) and Kelly et al. (2023): the graph topology determines which firms’ shocks co-propagate, making the channel of transmission explicit rather than latent.

To interpret how the GNN estimates conditional mean returns, we conduct a classical ANOVA test on whether firms have significantly distinct characteristics that are embedded by the GNN model. The results show that graph structural information such as node degree in the supply chain as well as firm-level characteristics like cash flow differ statistically significantly across firms (Table 7, Figure 9). This indicates that in addition to firm characteristics, the GNN embeddings also capture supply chain structure, with differences in degree centrality reflected in the learned representations that are used for conditional mean return estimation.

We design a Monte Carlo-based experiment to assess how robust the GNN model is to data missingness, considering that supply chain data are most likely not encompassing all actual business relationships of firms. For each model, we split firms based on their degrees and sizes in the SCG and measure the sensitivity of the model in separate strata. With random removal of a supply chain link, we find that the conditional mean return of the most peripheral firms estimated by the recommended ML model ($d = 3$) has variations that are 77% to 85% lower than one-hop ($d = 1$) and two-hop ($d = 2$) architectures respectively; for the most central firms, the $d = 3$ model demonstrates variations 77% to 80% lower than $d = 1$ and $d = 2$ models. The $d = 3$ model reduces sensitivity by 83% relative to $d = 2$ while concentrating remaining sensitivity in high-degree nodes, consistent with a network-propagation interpretation of the sensitivity gradient. By contrast, pricing sensitivity varies far less across size groups than across degree groups, suggesting that the GNN captures genuine graph structure rather than merely proxying for firm size in explaining cross-sectional returns.

1.1 Relation to Existing Literature

A growing literature documents that stock returns are predictable using other firms’ signals. Early machine learning models (Gu et al., 2020; Chen et al., 2024) focused exclusively on own-stock characteristics. Didisheim et al. (2024) provide a theoretical rationale for this approach, showing that in high-complexity settings cross-stock effects on the stochastic discount factor are negligible. Kelly et al. (2025) demonstrate that models explicitly exploiting cross-stock information achieve substantially higher out-of-sample predictive performance, and Kelly et al. (2023) show that even linear cross-predictability effects are economically large. This literature has established the empirical significance of cross-predictability but has not provide a systematic mechanism to explain its source. Our paper fills this gap by showing that supply chain topology determines which firms’ shocks co-propagate, providing the economic mechanism that the predictive evidence implies but does not explain.

[Herskovic \(2018\)](#) examines how production networks and supply chain dynamics generate systematic risks that impact asset returns. Their theoretical model predicts that the graph centrality and sparsity of supply chain networks are key determinants of return spreads.⁶ Their results rely on a structural model, and show that disruptions often lead to volatility clusters across firms within the same supply network. Our approach is fundamentally different, because we do not impose a structural relation between returns and supply-chain connections. Instead, we use a data-driven procedure that adaptively captures nonlinear network effects and aggregates extensive firm-level information. Our sensitivity analysis further examines the impact of degree centrality on expected returns cross-sectionally.

[Ding et al. \(2021\)](#) reports the cross-section of stock returns reaction to COVID-19 cases in fact depend on characteristics. Specifically, pandemic-related return drops vary as they condition on firm characteristics such as firm size, firm finances (credit and profit levels) as well as supply chain placements. We leverage this finding, but shift the focus from exogenous shocks to the temporal variations in supply chain linkages, analyzed in conjunction with panels of firm characteristics.

A growing literature documents supply chain relationships as material predictors of financial performance. [Wu \(2024\)](#) proposes a text-based metric for assessing firms’ exposure to supply chain risk, which captures how disruptions reverberate across firms with shared suppliers or customers. [Agca et al. \(2022\)](#) examine credit shock propagation within supply chains, showing that financial constraints in upstream firms can cascade down the supply chain, affecting downstream firms’ stock returns. This finding supports the argument that understanding supply chain structures is necessary for asset pricing, as these relationships carry material risk information. [Agrawal and Osadchiy \(2024\)](#) extend this perspective by analyzing inventory productivity within supply chains, showing how firm-level operational data can predict stock returns. We extend this literature by introducing a GNN architecture that learns from any observed subset of the supply chain network without pre-specifying link directions or neighborhood structures. The method adapts to the local topology, and is robust to the partial observability that characterizes supply chain data in practice.

[Chen et al. \(2024\)](#) show that deep learning frameworks provide the flexibility and scalability necessary for high-dimensional asset pricing datasets. We build on this by adapting a model specifically designed to handle relational data, enabling the GNN to exploit supply chain graph structure that standard deep learning architectures discard.

A few studies examine the theoretical and empirical properties of complex machine learning models in asset pricing. [Kelly et al. \(2022\)](#) study market return predictability and [Kelly et al. \(2024b\)](#) analyze futures return prediction; both demonstrate that complex models exhibit superior theoretical properties—partly due to the double-descent phenomenon in statistical learning ([Hastie et al., 2022](#))—and deliver significant improvements in market-timing Sharpe ratios. In a

⁶Some studies have analyzed the spillover effects of supply chain disruption. [Carvalho et al. \(2021\)](#) leverage the exogenous and regional nature of the East Japan earthquake shock to demonstrate the significant second-order effects of supply chain disruptions, which impact not only direct suppliers and customers of disaster-hit firms but also their indirect counterparts. Similarly, [Barrot and Sauvagnat \(2016\)](#) provide evidence of how idiosyncratic shocks propagate along supply chains, influencing the valuations of firms both directly and indirectly exposed to the initial shock.

parallel development focused on the cross-section, [Didisheim et al. \(2024\)](#) introduce artificial intelligence pricing theory and show that large factor models with far more factors than observations significantly outperform simpler benchmark models out-of-sample.

[Kelly et al. \(2024a\)](#) analyze deep neural networks trained by gradient descent. [Kelly et al. \(2025\)](#) go further, embedding a transformer directly in the stochastic discount factor to enable cross-asset information sharing via attention mechanisms. They show that exploiting cross-asset structure yields significant reductions in pricing errors relative to own-signal models. We complement this literature by identifying the structural source of cross-asset predictability: the supply chain network. Its graph topology determines which firms’ shocks co-propagate and therefore which cross-asset return dependencies are forecastable. Where transformer models learn cross-asset dependencies from data-driven co-movement patterns, our GNN anchors them in customer–supplier linkages, making the channel of transmission explicit rather than latent.

Another distinction between our study and the ones surveyed above concerns the choice of learning objective. [Chen et al. \(2024\)](#) minimize pricing errors directly, while [Didisheim et al. \(2024\)](#), [Kelly et al. \(2024a\)](#), and [Kelly et al. \(2025\)](#) employ the Maximal Sharpe Ratio Regression (MSRR) objective, training models to approximate the conditionally mean-variance efficient portfolio. Our framework follows [Gu et al. \(2020\)](#) in targeting conditional mean estimation, focusing on return predictability rather than portfolio optimization. This choice reflects our goal of isolating the supply chain channel as a source of expected return variation, independently of any particular portfolio construction criterion. Our method is designed to jointly learn from supply chain graph data and the panel of returns and firm characteristics, operating in a finite-sample setting. We provide a directly interpretable complexity parameter d that governs the radius of the local subgraph over which the model integrates. Larger d enables the model to aggregate information from more distant supply chain neighbors, improving predictive performance, but only up to a point. Beyond a critical threshold, performance deteriorates due to the well-documented oversmoothing phenomenon ([Chen et al., 2020](#)), whereby repeated graph convolutions cause node representations to converge and lose discriminative content. Models at excessive complexity also exhibit higher estimation variance. The optimal complexity in graph-panel joint learning is therefore bounded from above by the oversmoothing threshold, not by training sample size alone, a constraint specific to the graph-learning setting that has no direct analogue in standard panel-data models. We analyze this phenomenon formally in §5.1.

[Hou \(2007\)](#) documents an industry lead-lag effect whereby large-firm returns predict small-firm returns within the same industry, driven by gradual information diffusion across firms sharing common economic exposures. [Croce et al. \(2023\)](#) derive a structural leading premium from network centrality in a long-run risk model, showing that firms occupying central positions in the supply chain network command higher expected returns as compensation for bearing upstream risk. Our framework complements both studies along two dimensions. While [Hou \(2007\)](#) identifies information diffusion within industry boundaries, our GNN captures cross-predictability through explicit and multi-hop buyer–supplier linkages that frequently span industries, providing a more granular

propagation channel. Our GNN model uses lagged firm characteristic information, some of which are lagged up to 6-month, thus we have limited identifications on the speed of information propagation in supply chains. Our finding that high-degree firms exhibit 55% higher pricing sensitivity to missing supply chain links is consistent with the centrality premium of [Croce et al. \(2023\)](#), but established through a data-driven architecture rather than imposed by a structural model.

The rest of the paper is organized as follows. Section 2 introduces the model and discusses the procedure to generate the asset pricing factors based on sorted portfolios. Section 3 discusses the data and the supply chain GNN empirical performance of estimating firm returns, with a wide range of benchmarks from neural networks to Leontief regressions. We discuss asset pricing results obtained from portfolios formed using the empirically estimated ML model in Section 4. We provide economic interpretations of the GNN-based asset pricing model in Section 5. We conclude in Section 6. Appendix A develops an alternative graph representation of the supply chain network. Appendix B elaborates on the Transformer architecture, while Appendix C provides further details on the empirical analysis. To ensure reproducibility, our code is available on GitHub at <https://github.com/agcappo/SupplyChainAssetPricing>.

Notations: Unless otherwise specified, we use lower-case a to denote a scalar, bold lowercase $\mathbf{a}$ to denote a vector and bold uppercase $\mathbf{A}$ to denote a matrix. Firms are indexed by subscript i and time is indexed by subscript t . For a firm-time varying variable a_{it} , we use $\mathbf{a}_t$ to denote the cross-sectional vector of the same variable at time t , and $\mathbf{a}_i$ to denote the time-series vector of the same variable for firm i .

2 Methodology

We present an empirical framework that links firms’ excess returns to the channels through which supply-chain risks propagate. First, we motivate the use of supply-chain graphs to test for a supply-chain risk premium and introduce graph notation to compactly represent complex inter-firm relationships and risk-transmission mechanisms. Second, we describe our model of the conditional mean of excess returns with a specialized graph neural network (GNN). The GNN detects supply-chain risk exposure when characteristic-managed risk factors vary along the supply-chain graph.

2.1 Network Adaptive view of Supply Chain Risks

We define supply-chain risk as the exposure a firm bears when shocks originating elsewhere in the supply chain network—through disruptions to cash flows, inventories, sales, or production capacity—propagate to affect its own returns via direct or indirect customer–supplier linkages. The transmission mechanism can be triggered by events such as natural disasters, or credit distress in an upstream counterparty, where an adverse shock at one node impairs the operational or financial

position of connected firms, generating forecastable return co-movement across the network. Conversely, a demand shock experienced by a major customer may propagate upstream, compressing suppliers’ output targets and curtailing their investment plans. These propagation channels generate cross-firm return co-movement whose structure is directly mapped by observed supply-chain linkages, making the channel empirically testable.

In practice, supply-chain risk manifests as comovement in firm characteristics (e.g. inventory accumulation, capital expenditures, and working-capital needs tend to co-move between suppliers and customers). This suggests conditioning on characteristics that capture production volatility and investment risk, such as the change in the inventory-to-assets ratio (ΔPI2A) (Lyandres et al., 2008; Freyberger et al., 2020). A central question, however, is whether such a characteristic set, however rich, can fully summarize supply-chain risk.

Given an observed supply-chain network, it is natural to hypothesize higher-order interactions among firms that are not directly linked but are connected via multi-hop paths. Such pairs can exhibit conditional dependencies that traditional models fail to capture, because it is a priori unclear at what path length—and with what attenuation—supply-chain shocks affect firms’ returns.

This guides our choice of an *adaptive* statistical-learning framework that embeds the supply-chain graph in the information set for conditional-mean estimation. Rather than enumerating all potential risk channels, we employ a message-passing GNN to estimate the latent conditional expectation of returns, jointly conditioning on firm characteristics and the graph structure. The GNN thereby learns propagation and interaction effects that are unobservable in the cross-section of characteristics alone.⁷

Our approach is complementary to the growing literature on machine learning for asset pricing. While recent advances have focused on modeling nonlinearities or selecting informative firm characteristics, we argue that the supply chain graph provides an orthogonal source of information. It either contributes to return prediction by capturing omitted but economically meaningful information not contained in the firm characteristics, or it improves the estimation of the latent signals already embedded in existing variables such as inventory, investment, or profitability. Our empirical results confirm that the inclusion of supply chain data enhances predictive performance and improves the economic interpretability of the resulting factors.

2.2 Formulating the Pricing Problem with Supply Chains

Our objective is to empirically test whether supply chains serve as a source of priced risks in excess returns. Specifically, as shocks propagate from suppliers to customers, we examine whether risk premia also transmit along supply chain relationships.

Investors may call for supply chain risk premia across firms through a wide variety of channels. That is, supply chain risks may be channeled among firms even if they are not directly linked but

⁷Treating the supply-chain graph as extra information in the conditional mean estimation of firm-level returns is related to augmenting the covariance structure with graph-based priors. The graph can be viewed as implicitly regularizing or enriching cross-firm covariance by encoding economic relationships. Our approach, however, does not impose an explicit covariance model; it learns higher-order dependencies through the GNN’s nonlinear architecture.

are merely sharing a supply chain node. To illustrate these possibilities, consider the following two examples where firms are not directly linked, but supply chain risks still could at least partially explain their risk premia covariance:

Example 2.2.1: firms A and B are not linked as supplier-customer but have firm C as a shared supplier. If a natural disaster hits C , A and B 's production will be affected, and investors may expect risk premia on firms A and B conditional on C 's stress.

Example 2.2.2: firms A and B are not linked as supplier-customer but their suppliers have a shared customer C , so when firm C defaults, both A and B 's suppliers can suffer cash flow strains, potentially reducing A and B 's productions. This may also be a reason for investors to expect risk premia on firms A and B conditional on C 's stress.

To potentially capture such a broad spectrum of supply chain risk transfers, we require methods that take advantage of higher-order observed supply chain relationships (i.e., a network of supply chain relationships) rather than only direct supplier-customer linkages. As opposed to event-based studies, we identify supply chain risks as the variations of risk factors that reflect different aspects of firm operations, conditional on an observed supply chain network.⁸

In our analysis, we use the following notation: at time t , let N_t denote the number of firms for which firm characteristic-managed portfolio risk factors are observed.⁹ The empirical asset pricing literature seeks to identify a conditional mean function for excess returns in period t using only information available by period t , without look-ahead bias. Let Y_{it} denote the excess return of firm i at time t , and let $\mathbf{x}_{it}$ denote its vector of characteristic-managed portfolio returns, such as those constructed in Freyberger et al. (2020). Specifically, for each underlying firm characteristic, firm i is assigned to a decile and the associated decile-portfolio return enters $\mathbf{x}_{it}$ as a potential risk factor. Both $\mathbf{x}_{it}$ and Y_{it} are observed at monthly frequency with appropriate lagging to avoid look-ahead bias. We use p to denote the dimension of $\mathbf{x}_{it}$, and $\mathbf{X}_t \in \mathbb{R}^{N_t \times p}$ to denote the row-wise cross-section of firm characteristic-managed portfolio risks at time t . Our statistical learning problem models the conditional mean of excess returns in settings where both p and N_t may be large.

Graphs are the natural mathematical object to efficiently describe higher-order relationships. We represent the entire supply chain network of *all firms* at time t as a directed graph.¹⁰ Using graph notation, firm i is node i , and the set of nodes available in the supply chain data at time t is V_t . Note that V_t is a subset of the return universe, since some firms with returns or characteristic-managed portfolio risk factors may not appear in supply chain disclosures, so $|V_t| < N_t$. We define a directed edge $(j, i) \in E_t$ when firm j is a supplier of firm i at time t . Throughout the paper, the

⁸We use characteristic-managed factors from Freyberger et al. (2020). These firm characteristics are rich and publicly available, including, for example, Lyandres et al. (2008)'s changes in property, plants, and equipment and inventory over lagged total assets, and Palazzo (2012)'s ratio of cash and short-term investments to total assets.

⁹We provide more details on the comprehensive universe of U.S. listed firms that we consider in the empirical section's §3.1.

¹⁰Locally, a pair of firms may appear in different supply chains; in our graph representation, we record directed supplier-customer links.

neighborhood of node i refers to its suppliers:

$$\mathcal{N}_t(i) = \{j \in V_t : (j, i) \in E_t\}.$$

Supply chain graph (SCG) at time t is thus expressed as $SCG_t = (V_t, E_t)$. In our asset pricing problem, we associate each node with the firm’s characteristic-managed portfolio risk factors, and only require a D_t^{Graph} -dimensional vector to jointly express the asset pricing and relationship data:

$$D_t^{\text{Graph}} = (p + 1) \times |V_t| + |E_t|. \quad (1)$$

The first additive term in (1) is due to each firm having $(p + 1)$ -dimensional information of returns and firm-level characteristic-managed portfolio risk factors, multiplied by $|V_t|$ firms on the graph. The second additive term represents directed supply-chain relationships. In Appendix A, we illustrate why the proposed graph representation is efficient.

2.3 Pricing Kernel: Graph Neural Network by Message Passing

Building on the notation above, we model expected returns with a graph-based pricing kernel implemented through message passing.¹¹ At each layer l , node i updates its state by aggregating signals from in-neighbors $j \in \mathcal{N}_t(i)$ (upstream suppliers) and combining them with its own state. For each firm i , the model therefore applies a nonlinear graph-conditioned transformation of characteristic-managed portfolio risk factors to estimate the conditional mean return in period t .

Our object of interest is the firm-level conditional mean of excess returns in period t , conditioned on historical information available at time t . Let $\mathcal{F}_t = \sigma(\mathbf{X}_t, SCG_t)$ denote the sigma-field generated by the cross-section of firm characteristic-managed portfolio risk factors and the observed directed supply chain graph. We define

$$\mu_{it} \equiv \mathbb{E}[Y_{it} \mid \mathcal{F}_t]. \quad (2)$$

Consistent with the lagging conventions described above, $\mathcal{F}_t$ contains only information available by period t , so the conditioning set remains free of look-ahead information.

We approximate μ_{it} with a graph-conditioned nonlinear function class parameterized by Θ :

$$\hat{\mu}_{it} = g_i(\mathbf{X}_t, SCG_t; \Theta). \quad (3)$$

The function g_i is implemented through message passing, so each firm-level estimate depends on own risk factors and on network-mediated spillovers from economically linked firms.

Let $h_{it}^{(0)} \in \mathbb{R}^q$ be the initial representation of firm i , constructed from $\mathbf{x}_{it}$, where q is the embedding dimension. For each layer $l = 1, \dots, d$, define

$$m_{it}^{(l)} = \text{AGG}\left(\left\{f_{\text{msg}}\left(h_{it}^{(l-1)}, h_{jt}^{(l-1)}\right) : j \in \mathcal{N}_t(i)\right\}\right), \quad (4)$$

¹¹GNNs are a rapidly evolving field; for an introduction, see [Hamilton \(2020\)](#).

$$h_{it}^{(l)} = f_{\text{upd}}\left(h_{it}^{(l-1)}, m_{it}^{(l)}\right). \quad (5)$$

Here $m_{it}^{(l)} \in \mathbb{R}^q$ and $h_{it}^{(l)} \in \mathbb{R}^q$ are column vectors; $f_{\text{msg}} : \mathbb{R}^q \times \mathbb{R}^q \rightarrow \mathbb{R}^q$, AGG maps a finite set of q -vectors into $\mathbb{R}^q$, and $f_{\text{upd}} : \mathbb{R}^q \times \mathbb{R}^q \rightarrow \mathbb{R}^q$.

Embedding (general d -layer case). For firm i at time t , the model's *embedding* is the last hidden-layer state

$$\mathbf{e}_{it}^{(d)} \equiv h_{it}^{(d)} \in \mathbb{R}^q. \quad (6)$$

The pricing kernel maps this embedding to a scalar expected-return estimate through a linear readout. After d layers, the conditional-mean estimator is

$$\hat{\mu}_{it} \equiv \hat{Y}_{it} = a^\top \mathbf{e}_{it}^{(d)} + b. \quad (7)$$

with $a \in \mathbb{R}^q$ and $b \in \mathbb{R}$.

This recursion induces endogenous cross-sectional dependence in expected returns: supplier shocks affect customers' representations through $m_{it}^{(l)}$, and these effects travel along progressively longer directed production paths as l increases. The estimator therefore captures nonlinear spillover channels excluded by characteristic-only predictive regressions.

Before turning to the fully nonlinear specification, we present a linear benchmark purely for illustration. The objective is to provide intuition for message passing: neighborhood aggregation introduces network information into firm-level forecasts layer by layer. This linear case is an expository device, not the model used in estimation.

Example 2.3.1 Linear one- and two-layer cases. For this illustration, suppose $h_{it}^{(l)} \in \mathbb{R}^q$ is a column vector and $W_1, W_2 \in \mathbb{R}^{q \times q}$. We consider a simple aggregator function that takes the average of firms within its neighborhood $\mathcal{N}_t(i)$:

$$\text{AGG} = \frac{1}{|\mathcal{N}_t(i)|} \sum_{j \in \mathcal{N}_t(i)} (\cdot). \quad (8)$$

We further define the average for i th firm at l th layer,

$$m_{it}^{(l)} = \frac{1}{|\mathcal{N}_t(i)|} \sum_{j \in \mathcal{N}_t(i)} h_{jt}^{(l-1)}. \quad (9)$$

The simplest way to combine the $h_{it}^{(l-1)}$ vector and the $m_{it}^{(l)}$ vector is with the linear update:

$$h_{it}^{(l)} = W_1 h_{it}^{(l-1)} + W_2 m_{it}^{(l)}. \quad (10)$$

One-layer case. With $l = 1$, the representation is

$$h_{it}^{(1)} = W_1 h_{it}^{(0)} + W_2 \left(\frac{1}{|\mathcal{N}_t(i)|} \sum_{j \in \mathcal{N}_t(i)} h_{jt}^{(0)} \right). \quad (11)$$

In this one-layer case, the embedding is $\mathbf{e}_{it}^{(1)} = h_{it}^{(1)}$. Economically, the one-layer specification is a network-augmented linear factor model: expected returns depend on firm i 's own characteristic-managed portfolio risk factors and on average supplier-side factors in its immediate production neighborhood. Relative to a standard linear predictive regression, this introduces first-order cross-firm dependence through direct supply-chain exposure. In asset-pricing terms, the specification allows compensation for immediate upstream fragility in addition to firm-specific fundamentals.

Two-layer case. Substituting the first-layer relation into the second layer yields

$$\begin{aligned} h_{it}^{(2)} &= W_1 h_{it}^{(1)} + W_2 \left(\frac{1}{|\mathcal{N}_t(i)|} \sum_{j \in \mathcal{N}_t(i)} h_{jt}^{(1)} \right) \\ &= W_1^2 h_{it}^{(0)} + W_1 W_2 \left(\frac{1}{|\mathcal{N}_t(i)|} \sum_{j \in \mathcal{N}_t(i)} h_{jt}^{(0)} \right) + W_2 W_1 \left(\frac{1}{|\mathcal{N}_t(i)|} \sum_{j \in \mathcal{N}_t(i)} h_{jt}^{(0)} \right) \\ &\quad + W_2^2 \left(\frac{1}{|\mathcal{N}_t(i)|} \sum_{j \in \mathcal{N}_t(i)} \frac{1}{|\mathcal{N}_t(j)|} \sum_{k \in \mathcal{N}_t(j)} h_{kt}^{(0)} \right). \end{aligned} \quad (12)$$

In this two-layer case, the embedding is $\mathbf{e}_{it}^{(2)} = h_{it}^{(2)}$. Every term in the expansion is a $q \times 1$ vector: W_1^2 , $W_1 W_2$, $W_2 W_1$, and W_2^2 are all $q \times q$, premultiplying averaged $q \times 1$ vectors. The two middle terms are kept separate because matrix multiplication is generally non-commutative, so $W_1 W_2 \neq W_2 W_1$ in general. Economically, the last term introduces second-order transmission: supplier shocks that first affect firm j can then affect firm i through j 's own suppliers. This is the neighbors-of-neighbors channel emphasized by production-network asset pricing literature, where indirect bottlenecks and financing stress may be priced even without a direct contractual link. The two-layer system therefore embeds both direct and indirect network effects, yielding richer cross-sectional dependence than one-layer models. When $\mathcal{N}_t(i) = \emptyset$ (or the graph is absent), network terms drop out and the model nests a standard linear predictive regression.

The d -layer linear extension follows the same recursion:

$$h_{it}^{(l)} = W_1 h_{it}^{(l-1)} + W_2 \left(\frac{1}{|\mathcal{N}_t(i)|} \sum_{j \in \mathcal{N}_t(i)} h_{jt}^{(l-1)} \right), \quad l = 1, 2, \dots, \quad (13)$$

with $h_{it}^{(0)}$ initialized from firm characteristic-managed portfolio risk factors, and $\hat{\mu}_{it}^{(d)} = a^\top \mathbf{e}_{it}^{(d)} + b$. Iterating this recursion makes clear that d controls the effective network radius: larger d incorporates progressively longer supplier-path exposures in expected returns.

The empirical model we estimate is more general than this linear benchmark. In our implementation, f_{msg} and f_{upd} are nonlinear and parameterized with a Transformer architecture, one of several admissible nonlinear candidates. We provide implementation details for the empirical objective, regularization, and optimization are given in §3.2.1. To summarize, the central object carried across layers is the firm-level embedding $\mathbf{e}_{it}^{(d)} = h_{it}^{(d)}$, and expected returns are obtained by

a linear projection of that last hidden-layer representation.

We next characterize how layer depth d maps into local subgraph size and, in turn, into the economic reach of network effects. The corresponding equations of the GNN architecture are reported in Appendix B.1.

Remark 1. *Joshi et al. (2025) establish a formal equivalence between Transformers and Graph Neural Networks (GNNs), showing that Transformers implement a special case of GNN with dense, all-to-all attention operating on a fully connected graph.¹² Hence, the transformer-based asset pricing model of Kelly et al. (2025), which applies self-attention directly to a cross-section of assets, can itself be interpreted as a dense GNN where all asset pairs are connected. Our setting is materially different: as shown in Figure 2, supply chain graphs are sparse, with the number of edges growing only modestly relative to the number of nodes and most firm pairs having no direct link. In this sparse regime, unrestricted all-to-all attention would implicitly impose connections that have no economic counterpart, whereas our GNN enforces message passing exclusively along observed supplier-customer directed edges. This architectural choice restricts message passing to observed supplier-customer edges, so the model learns from statistical associations along the channels through which shocks may propagate, rather than implicitly assuming connections among all firm pairs. Layer depth then determines the radius of network information aggregation in a way that respects the observed graph topology.*

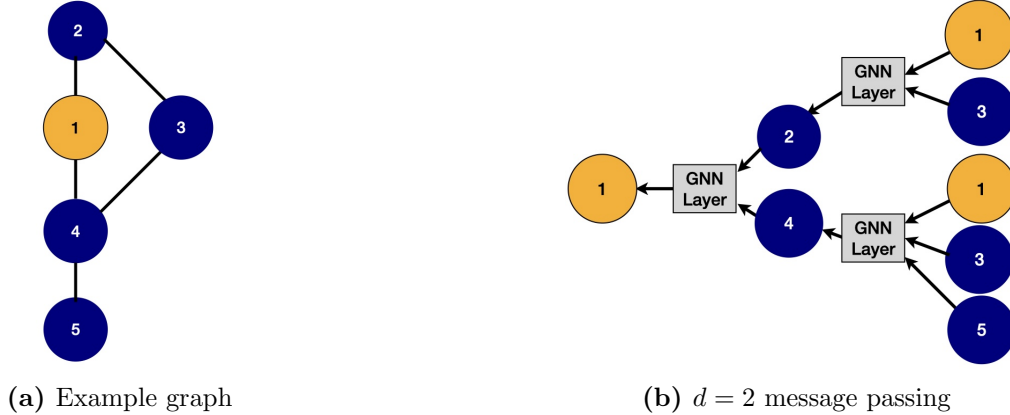
2.4 Supply Chain Subgraph

The depth of the GNN, represented by the number d of layers, determines the size of each node’s local subgraph that the model incorporates when updating node features. Each additional layer in the GNN allows information to propagate further across the graph, thereby increasing the number of nodes whose information influences a given target node. With d layers, each node receives messages from all nodes within d -hops (number of edges required to traverse) in the graph, effectively expanding its scope of information aggregation.

In a message-passing GNN, each node’s representation is iteratively updated by aggregating information from its neighbors. The number of GNN layers determines the number of “hops” each node’s representation can access, effectively defining the size of the local subgraph that each node captures. In Figure 1, we provide a visual illustration of how the message-passing process works, and highlight the connection between layer depth and subgraph size. Subfigure (a) illustrates a simple graph where node 1, the target node, is connected—either directly or indirectly—to nodes 2, 3, 4, and 5, which collectively form the local subgraph influencing node 1’s representation. Subfigure (b) uses arrows to indicate the directions of message passing within this subgraph. For this example, we demonstrate the GNN’s message-passing process across two layers ($d = 2$). In the first layer, node 1 aggregates information from its immediate (one-hop) neighbors, nodes 2 and 4, updating

¹²The self-attention mechanism computes pairwise importance weights between all tokens, which is mathematically equivalent to the attention aggregation performed by Graph Attention Networks, with the main difference that Transformers attend globally over all tokens rather than over a locally defined neighborhood.

Figure 1: The role of the parameter d in message passing controls the local subgraph size



its representation based on directly connected nodes. In the second layer, node 1 can now receive information from nodes two hops away—nodes 3 and 5—via nodes 2 and 4. This expanded reach allows node 1’s representation to reflect a broader local subgraph.

We also want to highlight that the message passed to the target node includes not only the characteristics of its neighbors but also those of the target node itself. Specifically in Figure 1, node 1, as the target node, receives messages from its immediate neighbors (nodes 2 and 4), which, in turn, embed node 1’s own characteristics during the first layer of message passing. This recursive structure means that a regression of the target node on its local context—a standard approach in traditional asset pricing—emerges naturally as a special case of the message-passing mechanism in GNNs when $d \geq 2$. The ability of the GNN to incorporate the target node’s features within the context of its broader supply chain relationships highlights the flexibility and generality of this framework, bridging conventional regression approaches and modern graph-based learning techniques. This property is central to the GNN’s ability to capture both direct and higher-order dependencies within the supply chain network.

The GNN’s layer depth controls how far each node can collect information within the graph. Multiple layers extend this to nodes further away, expanding each node’s receptive field within the graph. Thus, the number of layers defines the size of the local subgraph influencing each node. More layers allow deeper neighborhood capture but can introduce noise if the extended neighborhood becomes less relevant to the target node.

In the SCG setting, the layer depth d is a directly interpretable parameter quantifying how information is aggregated across supplier-customer relationships. Too small a value omits priced higher-order linkages: at $d = 1$, the model sees only direct counterparties and misses the indirect upstream pathways through which multi-tier disruptions propagate to the focal firm. Too large a value triggers oversmoothing (Chen et al., 2020): once d grows large enough that each firm’s aggregation neighborhood spans most of the supply chain graph, repeated aggregation drives firm embeddings toward indistinguishability, destroying the cross-sectional dispersion in expected returns that the model must preserve to generate return predictions. Optimal d balances these op-

posing forces—aggregating enough higher-order network structure to price multi-tier transmission risk while maintaining sufficient embedding heterogeneity to distinguish firms’ return exposures. We identify $d = 3$ as this optimum in §3.

3 Empirical Supply Chain Asset Pricing

We apply our methodology to investigate how supply chain data and firm characteristics explain expected returns. We present the sources and summary statistics of our data, including firm characteristics and supply chain relationships. In our empirical analysis, we begin by comparing the firm-level return estimation performance of the GNN model against benchmark models. Finally, we analyze the Sharpe ratios of portfolios formed using supply chain asset pricing factors.

3.1 Data

We combine firm-level financial data with supply chain relationship data. Our dataset captures both firm-specific and relational information essential for understanding the impact of supply chain structures on asset pricing.

3.1.1 Firm-Level Characteristics Data

We compute monthly stock excess returns Y_{it} and lagged characteristic-managed portfolio returns $\mathbf{x}_{it}$ from publicly listed firms on the NYSE, Nasdaq, and NYSE American (formerly known as the American Stock Exchange). The time period is from January 2003 to January 2024, providing a longitudinal view of firm performance across varying market conditions. The characteristic-managed risk factors $\mathbf{x}_{it}$ are built using $p = 64$ lagged financial indicators defined by Freyberger et al. (2020). The firm characteristics cover 6 categories of past return, investment, profitability, intangibles, value, and trading frictions, which overlaps with other well-known studies on “factor zoo” (Kozak et al., 2020; Hou et al., 2020).¹³ Following the literature’s practice (Kelly et al., 2019; Freyberger et al., 2020), we normalize the raw firm characteristics using the cross-sectional rank divided by the total number of firms N_t at time t , instead of directly using the values of the financial indicators. This cross-sectional rank normalization ensures that the raw firm characteristics fall within the bounded range of $[0, 1]$.

The variable Y_{it} denotes the realized monthly excess return for firm i at time t , computed by subtracting the risk-free rate retrieved from Kenneth French’s website. The vector $\mathbf{x}_{it}$ consists of characteristic-managed portfolio returns that summarize the firm’s financial position: for each of the 64 firm characteristics, we assign firm i to a decile and record the corresponding decile-portfolio return at time t , yielding a 64-dimensional snapshot of cross-sectional characteristic exposures. Both Y_{it} and $\mathbf{x}_{it}$ are constructed from the Center for Research in Security Prices (CRSP) and Compustat Fundamentals databases, accessed via Wharton Research Data Services (WRDS).

¹³For more details, see Freyberger et al. (2020)’s Table 1 and Internet Appendix.

Many of our $\mathbf{x}_{it}$ ’s are not originally defined at a monthly frequency, as many accounting characteristics are disclosed on a quarterly or annual basis. Therefore, we lag them using common practice in the literature (Gu et al., 2020) to ensure there is no look-ahead bias: we assume that monthly characteristics are delayed by at most 1 month, quarterly with at least 4 months lag, and annually with at least 6 months lag. Therefore, for the learning task of explaining excess returns at month t , we use the most recent monthly characteristics at the end of month $t - 1$, the most recent quarterly data by the end of month $t - 5$, and the most recent annual data by the end of month $t - 7$.

3.1.2 Supply Chain Relationship Data

To represent supply chain relationships, we use the supply chain graph SCG_t , where each node corresponds to a firm and each edge represents a supply chain linkage between firms. We leverage the FactSet Revere dataset as a more comprehensive data source of supply chain relationships. FactSet Revere hand-collects and verifies supply chain relationship information using various sources: 10-K filings (SEC’s SFAS 131 reporting standard), conference call transcripts, presentations to investors, company press releases, company websites, and major news-media reports.

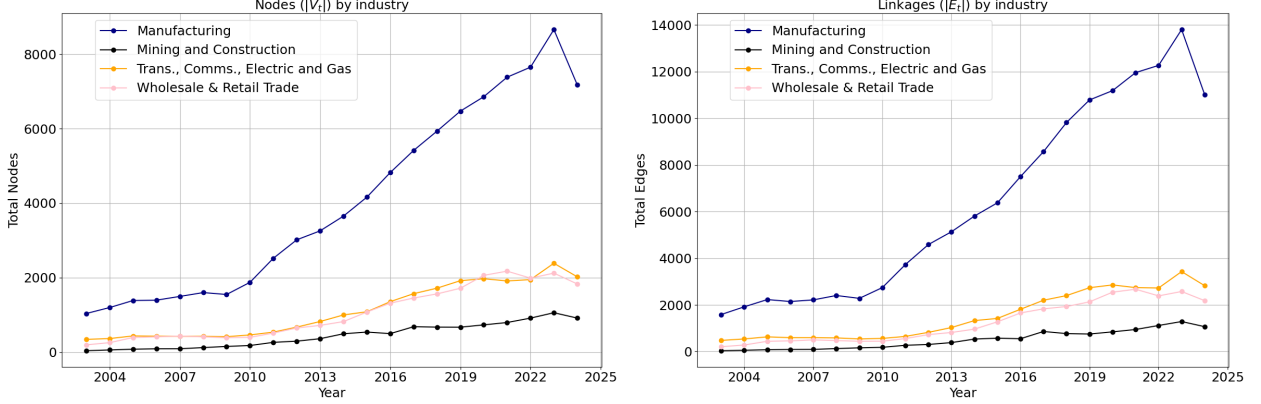
The retrieved SCG_t from the FactSet Revere relationships database begins in January 2003 and ends in June 2025. However, since the previously described characteristics data stops at January 2024, the intersection set contains data up to and including January 2024. In total, there are 1346 unique firms present in our constructed SCG, which have both supply chain relationships and firm characteristics data. When counting the firms that appear as counter-parties to those with available characteristics, our SCG has a total of 34,718 nodes. We do not include firms for which there was no available supply chain relationship data.

Supply chain disclosures are annual, whereas firm characteristics and returns are observed at monthly or quarterly frequency. We address this temporal misalignment by holding each year’s supply chain relationships fixed across all months within that year, integrating the annual graph structure into the monthly modeling framework. The lagging convention follows that of other annual financial data described in §3.1.1: to explain returns at month t , we use the most recently disclosed supply chain data available by the end of month $t - 7$. This ensures that supply chain information is strictly predetermined relative to the returns being forecast, supporting its interpretation as a predictive signal rather than a contemporaneous control.

To capture higher-order relationships within the supply chain network, we retain unlisted firms—those not traded on NYSE, NASDAQ, or NYSE American—in the graph. Excluding them would sever multi-hop paths connecting listed firms through shared unlisted intermediaries, truncating the message-passing routes through which indirect supply chain dependencies are transmitted. Retaining unlisted nodes ensures the model has access to the full relational structure of the network, not only its publicly traded subgraph.

Figure 2 summarizes the evolution of the SCG over our sample period, revealing three salient features: steady growth in both the number of firms and supply chain links, persistent sparsity

Figure 2: The number of nodes and edges by year.



This figure depicts the number of firms $|V_t|$ in a given industry, and the total number of edges $|E_t|$ at time t connected to firms in a given industry in our observed supply chain graph data. The data is annual frequency. We identify a firm’s industry by its Standard Industrial Classification (SIC) code.

(mean degree ranging from 0.9 to 1.7), and substantial heterogeneity across industries. The low mean degree constrains the effective aggregation horizon of the GNN and directly motivates the $d = 3$ design choice, as discussed in §5.1. Table A.2 provides granular statistics for the two most represented industries—Manufacturing and Wholesale & Retail Trade—reporting firm counts, link counts, and average degree by year.

The steady growth in observed links has a direct implication for our sample split. The training period spans 2003–2016 and the test period 2017–2024, a partition chosen so that the two subsamples contain approximately equal numbers of supply chain relationships—ensuring that out-of-sample evaluation is not conducted on a structurally denser graph than the one the model was trained on.

The SCGs are characterized by significant sparsity: each firm is typically connected to only a handful of direct suppliers or customers, with average degree ranging from 0.9 to 1.7 across industries and years (Table A.2). In 2023, for instance, the Manufacturing sector comprises approximately 9,000 firms and 13,900 supply chain edges, yielding an average degree of 1.6. This sparsity has a direct bearing on GNN design: because neighborhoods expand slowly with depth in a sparse graph, the model can reach multi-tier supply chain relationships at moderate depth without immediately encompassing the entire network. It also means that oversmoothing—rather than irrelevant signal aggregation from distant nodes—is the binding constraint on d , since embeddings converge once the aggregation horizon covers a large fraction of each firm’s connected component.

The SCGs also exhibit substantial cross-sectional heterogeneity across industries. Manufacturing dominates by node count (Figure 2), followed by Wholesale & Retail Trade, consistent with manufacturing firms’ roles as network hubs with dense buyer–supplier connections. Average degree varies systematically across sectors (Table A.2), with Manufacturing firms exhibiting higher connectivity than Wholesale & Retail Trade firms. We note, however, that supply chain disclosure requirements are uneven across sectors: Wholesale & Retail Trade relationships are likely under-

reported in our data (Agrawal and Osadchiy, 2024; Agca et al., 2022), introducing measurement noise that our model must accommodate. The GNN’s nonlinear, characteristic-conditioned aggregation is designed to extract cross-sectional return variation even when the observed graph is an incomplete image of the true network.

3.2 Conditional Excess Returns Mean with GNN

A central task in asset pricing is to characterize the conditional mean of excess returns, $\mathbb{E}[Y_{it} | \mathcal{F}_t]$, where $\mathcal{F}_t$ contains firm characteristics and economically driven inter-firm linkages observable by period t only. Conventional linear factor specifications map a small set of characteristics into expected returns, but they are not designed to capture dependence induced by supply-chain networks or higher-order interactions among firms (cf. Kelly et al., 2019; Freyberger et al., 2020). In our setting, shocks that originate upstream or downstream can propagate across multiple hops, so the relevant conditioning information is inherently relational and time-varying.

Graph neural networks (GNNs) provide a parsimonious yet flexible framework to incorporate this structure. Through message passing, a GNN aggregates and transforms information from a node’s neighborhood, allowing the conditional mean to reflect both a firm’s own characteristics and its position within the supply-chain graph. Compared with purely cross-sectional models, this approach can capture spillovers and non-linear interactions that arise from higher-order network paths, while remaining estimable at scale (Gu et al., 2020; Chen et al., 2024). In addition, by conditioning on a strong characteristics-only benchmark, we can isolate the incremental explanatory content of supply-chain linkages beyond what is already embedded in standard signals (Lettau and Pelger, 2020).

In what follows, we formalize our implementation. We specify the learning objective for the conditional mean, describe how firm-level covariates and the supply-chain graph enter the model via message passing, and detail the regularization and optimization choices that ensure out-of-sample robustness.

3.2.1 Estimation

In this subsection, we refer to the subset of data from January 2003 to December 2016 as the training dataset, denoted by $\mathcal{D}_{\text{train}}$, and the data from January 2017 to January 2024 as the test dataset, denoted by $\mathcal{D}_{\text{test}}$. The training of our model is based on minimizing the mean squared error (MSE) between estimated and actual returns.¹⁴

We shuffle the data during training to expose the model to a mixed sample of firms and time

¹⁴We conduct a comprehensive grid search to determine the optimal hyperparameters for our neural network model by minimizing MSE. Grids are constructed as a cartesian product of each parameter’s individual choice set. The hyperparameters and grids we search are: epochs - [10, 50, 80], dropout rate - [0.2, 0.3, 0.5, 0.8], attention heads - [1, 2, 3], output layer dimension - [40, 50, 60]. The selected parameters are as follows: 80 epochs, a dropout rate of 0.3 to control overfitting, an output layer dimension of 40, and 2 attention heads. We also have a learning rate of 0.001 for the Adam optimizer (other parameters of Adam set to the default values in pytorch), and an ℓ_2 regularization with scalar of 0.0001.

periods. Our shuffling strategy ensures that each mini-batch contains data from different periods and firms, allowing the model to learn generalized patterns across the full range of conditions within $\mathcal{D}_{\text{train}}$. By doing so, the model’s gradient descent algorithm is less likely to overfit to specific time periods or firm-specific trends and is instead encouraged to capture the broader structural patterns in the supply chain data.

In the estimation process, we ensure that the model has reached convergence before proceeding to evaluation. Convergence is verified by monitoring the training loss as a function of the number of batches processed. Specifically, we plot the training loss on the y-axis against the number of batches on the x-axis, with each line in the plot representing a different model configuration.

As shown in Figure A.3, each model configuration exhibits a declining training loss over the initial batches, eventually stabilizing as the model converges. The different colors represent various model configurations, allowing us to compare their convergence behaviors and select the configuration that best balances training efficiency and loss minimization. This plot provides a visual confirmation that the training process has stabilized, as indicated by the loss values plateauing over the final epochs. Convergence is observed when the training loss remains relatively constant across successive batches, suggesting that the model has minimized the objective function sufficiently given the data and selected hyperparameters.

3.2.2 Benchmarks

How much of the performance gain can be attributed to each component of our proposed return-explanation model? Our framework differs from prior empirical asset pricing studies in two ways: (i) we incorporate new supply chain graph data, and (ii) we introduce a GNN architecture in place of traditional linear models. This leads to two natural questions: First, do higher-order relationships in supply chain graphs improve return predictability? Second, conditional on using supply chain data, how much additional improvement comes from our GNN relative to standard linear benchmarks?

To showcase each source of improved estimation, we carefully construct benchmark comparisons by utilizing several well-established factor models in asset pricing:

- **Neural Network (NN):** A standard feed-forward neural network that provides a nonlinear benchmark without incorporating supply chain graph structure. The NN architecture maps firm characteristics-managed portfolio returns $\mathbf{x}_{it} \in \mathbb{R}^K$ to predicted excess returns through L hidden layers with ReLU activation functions $\sigma(\cdot) = \max(0, \cdot)$. For a network with L hidden layers, the prediction is given by:

$$\hat{Y}_{it}^{\text{NN}} = \mathbf{w}^{(L+1)\top} \sigma \left(\mathbf{w}^{(L)\top} \sigma \left(\dots \sigma \left(\mathbf{w}^{(1)\top} \mathbf{x}_{it} + b^{(1)} \right) \dots \right) + b^{(L)} \right) + b^{(L+1)}, \quad (14)$$

where $\mathbf{w}^{(\ell)}$ and $b^{(\ell)}$ denote the weight matrices and bias terms for layer ℓ , respectively. We evaluate NN architectures with $L \in \{2, 3\}$ hidden layers, where each hidden layer contains 40 neurons. Model parameters are estimated by minimizing the mean squared error loss

over the training set $\mathcal{D}_{\text{train}}$ using stochastic gradient descent with Adam optimization, with learning rate selected via grid search and early stopping based on validation set performance to prevent overfitting. This benchmark isolates the contribution of graph structure from the benefits of nonlinear modeling, allowing us to assess whether the GNN’s performance stems from its ability to leverage network topology or simply from its nonlinear parameterization.

- **Fama-French Five-Factor Model (FF5):** [Fama and French \(2015a\)](#) propose a widely accepted model that includes market, size, value, profitability, and investment factors.
- **Principal Component Analysis (PCA):** A factor model that captures risk premia through principal components of the empirical covariance matrix of firm characteristic-managed portfolio returns. We follow the procedures laid out in previous literature on PCA-based regression ([Lettau and Pelger, 2020](#); [Giglio and Xiu, 2021](#)), and provide detailed mathematical construction of this benchmark in Appendix C.1.
- **Neighboring Characteristics (NC)** This model takes the arithmetic average of the characteristics of a firm’s neighbors and then estimates a linear regression using these averages. Formally, for firm i at time t , define the neighbor-average characteristic vector:

$$\bar{\mathbf{x}}_{it} = \frac{1}{|\mathcal{N}_t(i)|} \sum_{j \in \mathcal{N}_t(i)} \mathbf{x}_{jt}. \quad (15)$$

We then estimate LASSO ([Tibshirani, 1996](#)) and Ridge ([Hoerl and Kennard, 1970](#)) regressions of excess returns on the concatenated vector $[\mathbf{x}_{it} \ \bar{\mathbf{x}}_{it}]^\top$:

$$\begin{aligned} \hat{\theta}^{\text{LASSO}} &= \operatorname{argmin}_{\theta} \mathbb{E}_{(i,t) \sim \mathcal{D}_{\text{train}}} \left[\left(Y_{it} - [\mathbf{x}_{it} \ \bar{\mathbf{x}}_{it}]^\top \theta \right)^2 \right] + \lambda_1 \|\theta\|_1, \\ \hat{\theta}^{\text{Ridge}} &= \operatorname{argmin}_{\theta} \mathbb{E}_{(i,t) \sim \mathcal{D}_{\text{train}}} \left[\left(Y_{it} - [\mathbf{x}_{it} \ \bar{\mathbf{x}}_{it}]^\top \theta \right)^2 \right] + \lambda_2 \|\theta\|_2^2, \end{aligned} \quad (16)$$

where the penalty hyperparameters λ_1 and λ_2 are selected from cross-validation.

We highlight the NC benchmark models as examples where we are using a simpler model than GNN, while conditioning on the supply chain graph information. If our machine learning approach produces better empirical performance, it would be evidence that widely used linear models are not flexible enough to capture the rich higher-order relationships that exist in supply chains to explain average returns.

We also use the NC benchmark model to generate NC asset pricing factors $\mathbf{F}_t^{\text{NC-LASSO}}$ and $\mathbf{F}_t^{\text{NC-Ridge}}$, in order to contrast against our supply chain asset pricing factors $\mathbf{F}_t$. To align NC with GNN model, we use the linear transformed concatenated characteristics-managed portfolio returns from either the estimated LASSO or Ridge:

$$\mathbf{x}_{it}^{\text{NC-LASSO}} = \hat{\theta}^{\text{LASSO}} \odot [\mathbf{x}_{it} \ \bar{\mathbf{x}}_{it}]^\top, \quad \mathbf{x}_{it}^{\text{NC-Ridge}} = \hat{\theta}^{\text{Ridge}} \odot [\mathbf{x}_{it} \ \bar{\mathbf{x}}_{it}]^\top, \quad (17)$$

where $\odot$ is the element-wise multiplication operator¹⁵. With $\mathbf{x}_{it}^{\text{NC-LASSO}}$ and $\mathbf{x}_{it}^{\text{NC-Ridge}}$, we can now treat them similarly as $\mathbf{x}_{it}$ to compute asset pricing factors $\mathbf{F}_t^{\text{NC-LASSO}}$ and $\mathbf{F}_t^{\text{NC-Ridge}}$. If the portfolios formed from these NC factors generate lower Sharpe ratios than our GNN-based factors, we have further evidence that the flexible model forms of GNN are better suited for capturing the rich relationships in the supply chain data.

3.2.3 Model Performance on Firm-Level Return Estimates

We now show the superior performance of the TransformerGNN, which jointly leverages SCG and firm characteristics data, as reported in Table 1. For firm-level monthly excess returns, we evaluate GNN models by comparing out-of-sample R^2 defined as:¹⁶

$$R^2 = 1 - \frac{\sum_{(i,t) \sim \mathcal{D}_{\text{test}}} (Y_{it} - \hat{Y}_{it})^2}{\sum_{(i,t) \sim \mathcal{D}_{\text{test}}} Y_{it}^2}, \quad (18)$$

where $\mathcal{D}_{\text{test}}$ refers to the out-of-sample period, Y_{it} is the actual excess return, and $\hat{Y}_{it}$ is the model’s predicted excess return.¹⁷ The R^2 measures the proportion of variance in excess returns that is explained by the model, with higher values indicating better predictive performance.

We compare our model against established benchmark models, including the zero constant prediction (ZERO), Fama-French Five-Factor model (FF5), Principal Component Analysis (PCA), two Neighboring Characteristics (NC) models that consider only 1-hop neighborhood in SCG and estimated with LASSO and Ridge, as well as standard multi-layer perceptron neural network (NN) benchmarks. We report out-of-sample performance metrics in terms of percentage improvement over FF5, as shown in Table 1. All models have reached convergence in our training, as reported in Figure A.3. We report all our results exclusively for the out-of-sample period of January 2017 through January 2024 to avoid look-ahead bias.¹⁸

First, we find direct empirical evidence that moderate graph depth improves firm-level return estimation. In Table 1, moving from $d = 1$ to $d = 3$ raises out-of-sample R^2 from 0.1493 to 0.1527, an absolute gain of 0.0034 (about 2.3% relative to $d = 1$). The $d = 4$ model delivers $R^2 = 0.1525$, which is nearly identical to $d = 3$. These patterns indicate that a one-hop architecture leaves

¹⁵For $\mathbf{x}_{it}^{\text{NC-LASSO}}$, we would have many zero entries due to LASSO’s sparsity.

¹⁶We note that in the R^2 metric (18), the denominator is the sum of the squared responses Y_{it}^2 *without demeaning*. In other words, we are benchmarking against a forecast of zero rather than the historical mean as in the statistical R^2 metric. As noted by Gu et al. (2020), predicting future excess stock returns with historical averages is not assumption-free, because the historical mean is estimated with significant noise, often performing worse than a forecast of zero.

¹⁷Throughout the paper, $\hat{Y}_{it}$ denotes the model’s estimated conditional mean of stock i ’s excess return in period t , formed exclusively from supply-chain and firm-characteristic information available at the portfolio formation date. No future return information enters the prediction at any stage; the timing convention Y_{it} (rather than $Y_{i,t+1}$) reflects that characteristics are observed at the beginning of period t and matched to returns realized during period t , consistent with standard practice in the empirical asset pricing literature (Gu et al., 2020).

¹⁸Our estimated GNN is estimated once given our train-test sample split, i.e. the out-of-sample results are based on a model estimated using pre-January 2017 data only (without re-estimation for subsequent periods) and evaluated on the test period. For completeness, in Appendix C.2, we present results of expanding window-fitted GNN, where we study the empirical performance of tangency portfolios constructed using a GNN model integrated with a NN model of firm characteristics-managed portfolio returns only, both of which are refitted over expanding windows.

Table 1: Out-of-sample predictive performance: GNN models versus benchmarks.

Model	Out-of-sample R^2		
	Value	Improvement vs FF5	Improvement vs NN3
<i>Panel A: Graph Neural Network Models</i>			
TransformerGNN ($d = 1$)	0.1493	+131.6%	+17.3%
TransformerGNN ($d = 2$)	0.1512	+132.0%	+18.8%
TransformerGNN ($d = 3$)	0.1527	+132.3%	+20.0%
TransformerGNN ($d = 4$)	0.1525	+132.3%	+19.8%
TransformerGNN ($d = 6$)	0.1253	+126.5%	-1.6%
TransformerGNN ($d = 10$)	-0.0030	+99.4%	-102.4%
<i>Panel B: Neural Network Benchmarks</i>			
NN2	0.1131	+124.0%	-11.2%
NN3	0.1273	+127.0%	Baseline
<i>Panel C: Linear Asset Pricing Benchmarks</i>			
FF5	-0.4722	Baseline	-470.9%
PCA ($k = 5$)	-0.4659	+1.3%	-466.0%
<i>Panel D: Neighborhood Linear Benchmarks</i>			
NC Ridge	0.0663	+114.0%	-47.9%
NC LASSO	0.0660	+114.0%	-48.1%

This table reports out-of-sample predictive R^2 for our GNN models and various benchmark models. We report the results above for the out-of-sample period of January 2017 through January 2024 to avoid look-ahead bias. Panel A presents TransformerGNN models with different neighborhood radii d , where d controls the number of message-passing layers. Panel B shows standard multi-layer perceptron neural network (NN) benchmarks without graph structure. Panel C includes linear asset pricing benchmarks: zero prediction (ZERO), Fama-French five-factor model (FF5), principal component analysis (PCA). Panel D presents neighboring characteristics (NC) models that use only first-hop supply chain neighbors with LASSO and Ridge regularization. The k parameter for PCA indicates the number of latent factors. Improvement percentages are calculated relative to FF5 (Panels A, C, D) and relative to the best-performing NN3 model of 3 layer Neural Network (Panel A).

economically relevant higher-order supply-chain structure unmodeled, while moderate additional depth improves fit.

Second, performance declines sharply when depth is increased excessively. As d rises from 3 to 10, out-of-sample R^2 falls from 0.1527 to -0.0030 , a drop of 0.1557. This pattern is consistent with a bias-variance trade-off and oversmoothing in deep message passing: when information is aggregated over too many hops, useful local signals are diluted by distant and noisier nodes. We provide a formal discussion of oversmoothing in §5.1.

Finally, the preferred TransformerGNN ($d = 3$) achieves out-of-sample $R^2 = 0.1527$, outperforming the best non-graph neural-network benchmark that uses the feedforward architecture (NN3, $R^2 = 0.1273$) as well as neighborhood linear models (NC Ridge, $R^2 = 0.0663$; NC LASSO, $R^2 = 0.0660$). These results indicate that incorporating supply-chain structure via message passing adds predictive content beyond flexible nonlinear models that omit the graph and beyond linear specifications that use only local neighbor summaries. Because asset pricing is not defined by

Table 2: Joint spanning regression of GNN predicted returns on the NN2, NN3, FF5, PCA, NC Ridge, NC LASSO benchmarks.

Specification	$\hat{a}_d$	$t(\hat{a}_d)$	R^2
GNN ($d = 1$)	0.0218	44.6751	0.8782
GNN ($d = 2$)	0.0197	49.6512	0.8661
GNN ($d = 3$)	0.0235	72.4248	0.8988
GNN ($d = 4$)	0.0268	86.0512	0.9018
GNN ($d = 6$)	0.0252	125.2749	0.8855
GNN ($d = 10$)	0.0013	348.6447	0.0007

This table reports results from the joint spanning regression of the depth- d GNN predicted return on all six non-graph benchmark predicted returns—NN2, NN3, FF5, PCA, NC Ridge, and NC LASSO—as specified in Equation (19). For each depth $d \in \{1, 2, 3, 4, 6, 10\}$, we report the estimated intercept $\hat{a}_d$, its Newey-West t -statistic (with lag length 12), and the regression R^2 . These regressions were estimated using exclusively the out-of-sample period of January 2017 through January 2024 to explore incremental explaining power of the GNN in the out-of-sample period. A positive and statistically significant intercept indicates that the GNN predicted return contains return-predictive information not spanned by any linear combination of the benchmark menu. The R^2 measures the fraction of GNN predicted return variation explained jointly by all non-graph predictors; the complement $(1 - R^2)$ represents the residual graph-specific component. Slope coefficients $\hat{\beta}_d$ are estimated but not tabulated, as their individual magnitudes are difficult to interpret economically when benchmark predicted returns are mutually correlated. All regressions are estimated over the out-of-sample period on firm-month observations. The dependent variable in each regression is $\hat{Y}_{it}^{\text{GNN}(d)}$.

predictive fit alone, we next study tradable-portfolio performance in §4.

Spanning relative to a non-graph NN benchmark. To assess whether the GNN captures return-predictive information not embedded in any linear combination of existing benchmarks, we regress each depth- d GNN predicted return jointly on the predicted returns from all non-graph models in our benchmark suite, including NN2, NN3, FF5, PCA, NC Ridge, and NC LASSO:

$$\hat{Y}_{it}^{\text{GNN}(d)} = a_d + \beta_d^\top \mathbf{z}_{it} + u_{it}, \quad (19)$$

where $\mathbf{z}_{it}$ stacks the predicted returns from all six non-graph benchmarks. This joint spanning regression is a strict test: if the GNN merely repackages information already present in characteristics-only or factor-based predictors, the intercept a_d would be indistinguishable from zero. We do not tabulate the slope coefficients $\hat{\beta}_d$, as their individual magnitudes are difficult to interpret economically when the benchmark predicted returns are mutually correlated. Instead, we focus on two economically interpretable objects: the intercept a_d and the Sharpe ratio of a long-minus-short portfolio sorted on the residualized predicted return $\hat{Y}_{it}^{\text{GNN}(d), \perp} \equiv \hat{Y}_{it}^{\text{GNN}(d)} - \hat{\beta}_d^\top \mathbf{z}_{it}$, which directly quantifies the tradable value of graph-conditioned information after the full benchmark menu has been projected out.

Table 2 reports the results. The intercepts are positive and strongly significant across all moderate-depth specifications. For $d = 3$, the joint regression yields $\hat{a}_3 \approx 2.4\%$ per month with a Newey-West t -statistic well above the [Chen and Zimmermann \(2022\)](#) recommended threshold of

$|t| = 3.0$, confirming that the supply-chain graph contributes a component of the predicted return that no linear combination of the six benchmark predictors can replicate. The uniformly high R^2 values for moderate d reflect shared variation driven by common characteristic inputs rather than a failure of spanning: all models target the same conditional mean of stock returns and draw on overlapping firm characteristics, rendering their predicted returns mechanically correlated. The economically meaningful quantity is the residual fraction $(1 - R^2) \approx 10\%$, which captures the variation in the GNN predicted return uniquely attributable to supply chain graph structure, and not reproducible by any non-graph predictor.

The $d = 10$ specification stands apart. Its intercept collapses to near zero and its R^2 falls to essentially zero—not because the benchmark suite successfully spans the GNN predicted return, but because the $d = 10$ GNN has lost virtually all return-predictive content, as confirmed by the near-zero out-of-sample R^2 in Table 1. The cause is *oversmoothing*, where repeated message passing drives all firm embeddings toward a common vector, eliminating the cross-sectional heterogeneity that generates return predictability. We investigate this mechanism systematically in §5.1, tracing embedding geometry, out-of-sample R^2 , and portfolio Sharpe ratios across depths to establish precisely where and why predictive content deteriorates.

3.3 Comparing GNNs with Leontief Network Regressions

To evaluate the benefits of our GNN approach, we compare it to a benchmark based on a Leontief inverse-style representation of the supply chain network. The modified Leontief regression predicts returns as linear functions of successive powers of the adjacency matrix, encoding higher-order interactions but imposing the same coefficient across all firms and network distances. This linear, homogeneous structure prevents adaptation to heterogeneity in firm characteristics or positions of the firm in the network. In contrast, the GNN is a nonlinear architecture that learns node-specific and feature-specific weights, enabling it to adaptively capture multi-hop relationships and richer propagation patterns than the Leontief framework.

3.3.1 Mathematical Setup

Firms’ returns of month t are $\mathbf{y}_t$. Let $\mathbf{X}_t \in \mathbb{R}^{N_t \times K}$ denote the matrix of firm characteristics or characteristic-managed portfolios for month t , with N_t firms and K characteristics. Let A_t denote the $|V_t| \times |V_t|$ supply chain adjacency matrix for the same month. The number of firms N_t may vary over time as firms enter or exit the market, and the adjacency matrix A_t captures the dynamic supply chain relationships among these firms.

We adapt Leontief (1986)’s input–output framework to the inter-firm setting, where a firm’s return depends not only on its own characteristics but also on shocks propagating through its direct and indirect supply chain links. Following the linear network framework of Acemoglu et al. (2012), these interactions can be represented by the Leontief inverse L_t of the adjacency matrix at time t ,

expressed as a convergent power series under a normalization factor ρ :

$$L_t(\rho) = (I - \rho A_t)^{-1} = \sum_{k=0}^{\infty} A_t^k \rho^k. \quad (20)$$

Here, I denotes the identity matrix and ρ is a scaling parameter. The series converges if and only if $\rho < 1/\lambda_{\max}(A_t)$, where $\lambda_{\max}(A_t)$ is the spectral radius of A_t , i.e., the spectral radius of ρA_t must be strictly less than one. The Leontief inverse captures the cumulative effect of shocks propagating through the network, allowing us to linearly model how a shock to one firm can affect others through their supply chain links. The linear effects of $\mathbf{X}_t$ with loading $\mathbf{b}$ can be expressed as:

$$\mathbf{y}_t = \sum_{k=0}^{\infty} A_t^k \rho^k \mathbf{X}_t \mathbf{b} + \epsilon_t, \quad (21)$$

where $\mathbf{b}$ is the loading parameter corresponding to $\mathbf{X}_t$ and ϵ_t is the idiosyncratic movements of returns at time t . The first-order effect A captures direct neighbors (suppliers or customers), the second-order effect A^2 captures neighbors-of-neighbors, and so on. This linear representation stands in contrast to the GNN, which can learn node-specific, edge-specific, and feature-specific weights in a nonlinear fashion through the message-passing architecture.

We estimate the parameters using a penalized regression:

$$\min_{\rho, \mathbf{b}} \sum_{t \in \mathcal{D}_{\text{train}}} \|\mathbf{y}_t - L_t(\rho) \mathbf{X}_t \mathbf{b}\|_2^2 + \lambda \|\mathbf{b}\|_1 \quad \text{s.t.} \quad 0 \leq \rho < \frac{1}{\lambda_{\max}(A_t)}, \quad (22)$$

where $\mathcal{D}_{\text{train}}$ denotes the training dataset, $\mathbf{y}_t$ is the $N_t \times 1$ vector of realized firm excess returns, and λ is the LASSO penalty parameter. The LASSO regularization serves two purposes: it controls overfitting and it performs implicit feature selection among the characteristic-managed portfolios.

Denote the estimated coefficient $\mathbf{b}$ by $\hat{\mathbf{b}}$. The estimated linear return equation is

$$\hat{\mathbf{y}}_t = \mathbf{X}_t \hat{\mathbf{b}}.$$

This specification serves as a direct linear benchmark for our GNN model. Economically, it permits higher-order network propagation but restricts all paths to decay at the same scalar rate ρ , while transforming features homogeneously across nodes. By contrast, the GNN applies heterogeneous, learned weights across nodes, neighbors, and firm characteristics.

3.3.2 Empirical Evaluation of Leontief Regression Benchmarks

Table 3 reports out-of-sample R^2 for the Leontief regression benchmark compared to our GNN model.

We further decompose the linear model's performance into contributions from successive network orders—first-order ($A_t \rho X_t b$), second-order ($A_t^2 \rho^2 X_t^2 b$), up to fifth-order interactions, so as to quantify how much predictive power stems from simple network propagation.

Table 3: Out-of-sample predictive performance: GNN models versus network regressions.

Model	Out-of-sample R^2	
	Value	Improvement vs Leontief
<i>Panel A: Graph Neural Network Models</i>		
TransformerGNN ($d = 1$)	0.1493	+71.0%
TransformerGNN ($d = 2$)	0.1512	+73.2%
TransformerGNN ($d = 3$)	0.1527	+74.9%
<i>Panel B: Leontief Network Regression</i>		
Full Leontief Regression	0.0873	Baseline
<i>Panel C: Partial Leontief Network Regression</i>		
First-Order	0.0165	-81.1%
Up to Second-Order	0.0229	-73.8%
Up to Third-Order	0.0284	-67.5%
Up to Fourth-Order	0.0336	-61.5%
Up to Fifth-Order	0.0382	-56.2%

This table reports out-of-sample predictive R^2 for GNN models and Leontief network regression benchmarks. We report all our results exclusively for the out-of-sample period of January 2017 through January 2024 to avoid look-ahead bias. Panel A presents TransformerGNN models with different neighborhood radii d , where d controls the number of message-passing layers. Panels B and C shows Leontief network regression models that incorporate network effects through successive powers of the adjacency matrix, with specifications ranging from first-order through fifth-order network terms. The Leontief regression imposes the same coefficient across all firms and network distances, while the GNN learns node-specific and feature-specific weights. Improvement percentages are calculated relative to the first-order Leontief regression baseline.

Empirically, an order-matched comparison strengthens the economic interpretation. At one hop, TransformerGNN with $d = 1$ achieves $R^2 = 0.1493$, versus 0.0165 for the first-order partial Leontief model. At two hops, TransformerGNN with $d = 2$ delivers $R^2 = 0.1512$, versus 0.0229 for the second-order partial Leontief specification. At three hops, TransformerGNN with $d = 3$ yields $R^2 = 0.1527$, versus 0.0284 for the third-order partial Leontief model. Holding network reach fixed in this way isolates functional-form differences: the GNN’s nonlinear, node-adaptive mapping captures heterogeneous transmission patterns that rigid linear polynomial propagation may have misspecified. The GNN at $d = 3$ also dominates the full fifth-order Leontief regression by a wide margin (0.1527 versus 0.0873), confirming that the incremental predictive content of supply-chain graph structure is not recoverable by extending the polynomial order of linear network aggregation.

4 Asset Pricing Interpretation

Having established the accuracy of the GNN framework in estimating conditional average returns, we now turn to financial economic tests and their linkage to interpretable supply chain structures. A crucial test for any new source of return predictability is whether it gives rise to a profitable, tradable investment strategy and whether the resulting pricing factor captures risks not spanned by existing models. To this end, we construct and evaluate asset pricing factors based

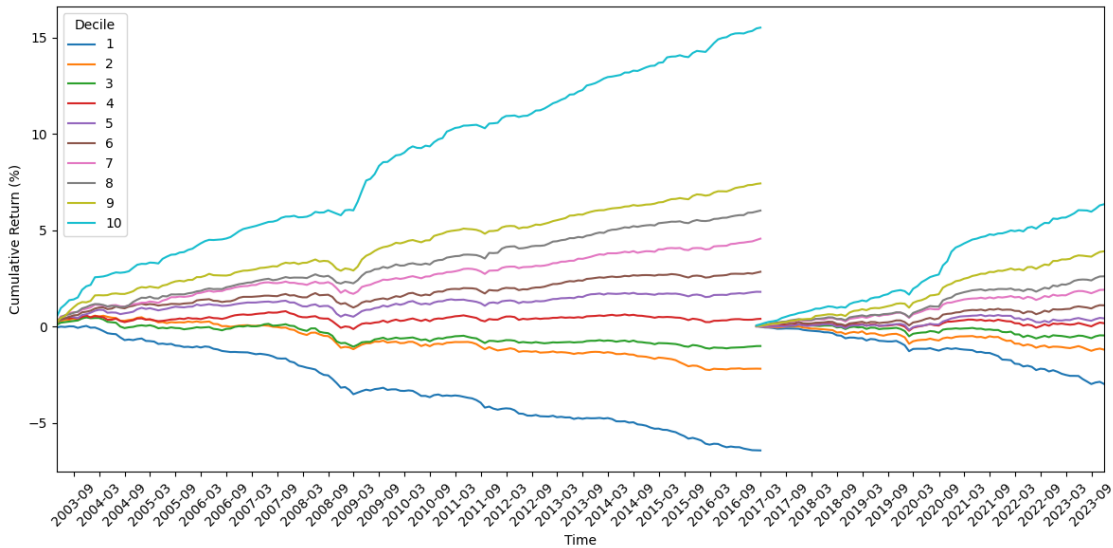
on different local graph sizes d , derived from the GNN’s excess return estimation and assess their relationship with established benchmarks.

4.1 Constructing a Tradable Factor from GNN Predictions

We translate the GNN’s firm-level return estimates into a tradable investment strategy using a standard portfolio-sorting methodology, providing a direct measure of the economic value of supply chain information in cross-sectional return prediction.

Decile Portfolio Performance. Each month, we sort firms into decile portfolios based on their GNN-predicted excess returns $\hat{Y}_{it}$ and track the equal-weighted realized return of each decile out-of-sample. Figure 3 plots the cumulative performance of these portfolios over the test period. The rank ordering is near-monotonic: cumulative returns increase systematically from Decile 1 through Decile 10, with the top decile substantially outperforming the bottom decile throughout the sample. This pattern indicates that the GNN’s predicted return rankings align with realized return rankings in the cross-section, a direct measure of the model’s ability to identify firms with persistently high or low expected returns.

Figure 3: Cumulative Returns of Decile-Sorted Portfolios



This figure plots both in- and out-of-sample cumulative returns of ten equal-weighted portfolios. The in-sample period ends on December 2016. The out-of-sample period has lines re-starting at 0 to visualize the persistent patterns of different deciles’ returns, and the out-of-sample period ranges from January 2017 through January 2024. Portfolios are formed monthly by sorting firms on their GNN-predicted excess returns. The model used is the TransformerGNN with a depth of $d = 3$.

Long-Short Factor Construction. The monotone return spread across deciles motivates a zero-investment long-minus-short factor, following standard practice in the machine learning asset pricing literature (Chen et al., 2024; Freyberger et al., 2020). At the end of each month t , all firms are ranked by their model-predicted return $\hat{Y}_{it}$ and assigned to ten deciles. Stocks are held in equal weights within each decile, and the portfolio is rebalanced monthly. The supply chain factor is the

Table 4: Out-of-sample Asset Pricing Performance of Supply Chain Factors

Model Name	Out-of-sample	
	ER	SR
GNN ($d = 1$)	0.0648	0.8562
GNN ($d = 2$)	0.0682	0.8988
GNN ($d = 3$)	0.1106	1.3923
GNN ($d = 4$)	0.1078	1.3914
GNN ($d = 10$)	-0.0003	-0.0095
NC Ridge	0.1156	1.1513
NC LASSO	0.1148	1.1946

This table reports the out-of-sample expected return (ER) and Sharpe ratio (SR) of long-minus-short portfolios managed by GNN (at various message-passing depths d), NC Ridge, and NC LASSO over the out-of-sample period $\mathcal{D}_{\text{test}}$. We report the results above for the out-of-sample period of January 2017 through January 2024 to avoid look-ahead bias. All portfolios are constructed as defined in Equation (23) (§4.1): firms are sorted monthly into deciles on model-predicted returns, with stocks equal-weighted within each decile.

return spread between the top and bottom decile portfolios,

$$F_t = R_{10,t} - R_{1,t}, \quad (23)$$

where $R_{j,t}$ is the equal-weighted return of decile j at time t . The same construction applies to all benchmark long-short portfolios reported in this paper, with the sorting key $\hat{Y}_{it}$ replaced by the respective model’s predicted return. Varying the message-passing depth d yields a family of supply chain factors. Table 4 reports the out-of-sample expected return and Sharpe ratio for $d \in \{1, 2, 3, 4, 10\}$ alongside the NC Ridge and NC LASSO benchmarks over the out-of-sample period $\mathcal{D}_{\text{test}}$.

Our supply chain-managed portfolios deliver out-of-sample Sharpe ratios and expected returns that materially exceed those of NC graph and firm-characteristic benchmark models. Table 4 reports the results, where each row corresponds to a distinct choice of supply chain sub-graph size d . Second and third-order network effects are necessary to price the cross-section: relative to the $d = 3$ specification, the $d = 1$ model delivers $\sim 40\%$ lower expected returns and Sharpe ratios, while $d = 2$ underperforms by $\sim 38\%$ and $\sim 35\%$, respectively. Limited network depth fails to capture the priced higher-order supply chain linkages that generate the factor’s return and efficiency advantages.

Nonlinear supply chain aggregation dominates its linear counterparts on a risk-adjusted basis. NC Ridge and NC LASSO deliver slightly higher expected returns than $d = 3$ GNN ($\sim 5\%$ and $\sim 4\%$, respectively), but at the cost of Sharpe ratios that are $\sim 17\%$ and $\sim 14\%$ lower—confirming that the GNN’s advantage lies in efficiency relative to linear characteristic benchmarks.

One might ask whether the declining Sharpe ratio beyond $d = 3$ reflects the standard bias-variance trade-off, whereby additional model complexity reduces bias at the cost of higher estimation variance. We argue it does not. In dense factor architectures of the kind studied by [Didisheim et al. \(2024\)](#), additional capacity can improve return predictability because richer parameterization

Table 5: Asset-pricing properties of GNN predicted returns orthogonalized to NN2, NN3, FF5, PCA, NC Ridge, NC LASSO benchmarks.

Residualized portfolio returns	Mean	Volatility	Sharpe Ratio
$\hat{Y}_t^{\text{GNN}(1),\perp}$	0.0525	0.0037	0.8660
$\hat{Y}_t^{\text{GNN}(2),\perp}$	0.0546	0.0034	0.9354
$\hat{Y}_t^{\text{GNN}(3),\perp}$	0.0633	0.0035	1.0682
$\hat{Y}_t^{\text{GNN}(4),\perp}$	0.0568	0.0039	0.9115
$\hat{Y}_t^{\text{GNN}(6),\perp}$	0.0431	0.0045	0.6422
$\hat{Y}_t^{\text{GNN}(10),\perp}$	0.0024	0.0010	0.0766

This table reports asset-pricing properties of GNN predicted returns orthogonalized to all six non-graph benchmarks (NN2, NN3, FF5, PCA, NC Ridge, NC LASSO). The residualized predicted return $\hat{Y}_{it}^{\text{GNN}(d),\perp} \equiv \hat{Y}_{it}^{\text{GNN}(d)} - \hat{\beta}_d^\top \mathbf{z}_{it}$ is constructed by removing the projection of the depth- d GNN predicted return onto the full benchmark menu, using the coefficient estimates from Equation (19). These regressions were estimated using exclusively the out-of-sample period of January 2017 through January 2024 to explore incremental explaining power of the GNN in the out-of-sample period. For each depth d , we report the mean and time-series volatility of the residualized predicted return, and the Sharpe ratio of a long-minus-short portfolio $\hat{Y}_t^{\text{GNN}(d),\perp}$ constructed following Equation (23), with the sorting key replaced by the residualized predicted return $\hat{Y}_{it}^{\text{GNN}(d),\perp}$. These statistics directly quantify the tradable economic value of the graph-conditioned component of the GNN predicted return that is orthogonal to all non-graph predictors (NN2, NN3, FF5, PCA) and the NC graph models. A Sharpe ratio that remains large after orthogonalization indicates that the supply-chain graph contributes priced information beyond what any linear combination of the benchmark models captures.

better approximates the true pricing function. GNNs operating on sparse graphs with local cyclic structure face a qualitatively different constraint: increasing message-passing depth causes neighborhood aggregation to homogenize firm embeddings, not merely overfit to noise. No additional training epochs or parameters can recover the firm-specific variation that repeated neighborhood aggregation has erased: once all embeddings collapse toward a common vector, the cross-section of expected returns becomes essentially flat, and the model’s return-predictive content disappears. The supply chain graph topology therefore imposes a structural complexity ceiling at $d = 3$, not a standard overfitting boundary that shrinkage or dropout can shift. We document this mechanism formally in §5.1.

To assess the asset pricing implications of the GNN’s incremental predicted returns beyond the benchmark menu, we use the residuals from Equation (19) to form a long-minus-short portfolio factor $\hat{Y}_{it}^{\text{GNN}(d),\perp}$ for each depth d , and report its mean return, volatility, and Sharpe ratio.

Table 5 quantifies the tradable value of the GNN’s graph-specific component. After projecting out all six benchmark predicted returns, a long-minus-short portfolio sorted on the residualized $d = 3$ GNN predicted return earns an out-of-sample Sharpe ratio of 1.07. Sharpe ratios rise from $d = 1$ (0.87) through the $d = 3$ peak (1.07), then decline monotonically through $d = 6$ (0.64) to $d = 10$ (0.08), consistent with the oversmoothing mechanism. The spanning evidence therefore establishes that the supply-chain GNN captures priced information orthogonal to the full non-graph benchmark menu, with incremental value concentrated at $d = 3$, where higher-order supply chain structure is captured without oversmoothing.

4.2 Regression Analysis with Test Portfolios

We assess whether the GNN-managed supply chain factor F_t carries independent pricing information beyond what established risk factors already explain. The test follows [Fama and French \(1993\)](#)’s time-series regression approach: for each test portfolio p , we regress its excess return on F_t and a set of control factors $\mathbf{z}_t$, then examine whether F_t carries a statistically significant loading. We conduct this test on three distinct sets of test assets, each designed to probe a different dimension of the return cross-section.

- **Fama–French 25 size-value portfolios (FF25).** These 25 portfolios are formed by independently sorting stocks into five size quintiles (based on market equity) and five book-to-market quintiles, then taking the intersection, following [Fama and French \(1993\)](#). They are the canonical test assets for models claiming to explain the value and size premia.
- **Fama–French 30 industry portfolios (FF30).** These 30 portfolios classify firms into broad industry groups based on four-digit SIC codes, following [Fama and French \(1997\)](#). They probe whether the factor explains return variation across the production structure of the economy, rather than within characteristic-sorted portfolios.
- **Jensen–Kelly–Pedersen 13 theme factors (JKP13).** [Jensen et al. \(2023\)](#) construct 13 long-minus-short characteristic-managed portfolios, each aggregating the return signal from a cluster of related firm-level anomalies into a single representative “theme” factor. The 13 themes span investment, debt issuance, seasonality, momentum, profitability, quality, low-risk, value, accruals, size, short-term reversal, idiosyncratic volatility, and skewness—collectively covering the major cross-sectional return anomalies documented in the literature.

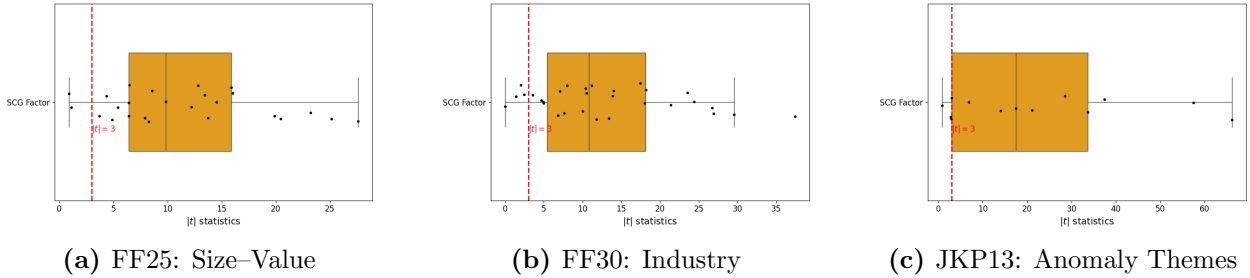
We estimate the regressions and compute the t -statistics with the entire data period, ranging from January 2003 through January 2024. All data are obtained from Kenneth R. French’s website (FF25, FF30) and from the authors’ website (JKP13). For each test portfolio, we run a monthly time-series regression of its excess return on the supply chain factor F_t and the five Fama–French control factors ([Fama and French, 2015b](#)) (market excess return, size, value, profitability, and investment), then examine the absolute t -statistic on the supply chain factor loading against the $|t| = 3$ threshold recommended by [Chen and Zimmermann \(2022\)](#) for robust inference under multiple testing.¹⁹

FF25: Size-value double sorts. Figure 4a summarizes the t -statistics for $\hat{\beta}_p$ across the 25 size-value sorted portfolios. Over 92% of portfolios (23 out of 25) exceed the $|t| = 3$ threshold, providing strong evidence that the supply chain factor carries pricing information beyond the five Fama–French factors across the entire size-value grid. The economic mechanism is that supply chain exposure is not uniform across size-value cells: small-growth firms—the bottom-left corner of

¹⁹We evaluate results on the out-of-sample period to assess whether the significance of F_t persists beyond the estimation sample, following the multiple-testing framework of [Barillas and Shanken \(2018\)](#); [Chen and Zimmermann \(2022\)](#); [Pelger and Zou \(2024\)](#).

the FF25 grid—tend to operate as upstream suppliers with concentrated customer relationships, making their returns highly sensitive to demand shocks that propagate through the production network (Cohen and Frazzini, 2008; Ding et al., 2021). Large-value firms, by contrast, tend to occupy downstream customer positions whose return variation is more fully captured by size and book-to-market exposures alone.

Figure 4: Factor Loadings on the Supply Chain Factor Across Three Test Asset Sets

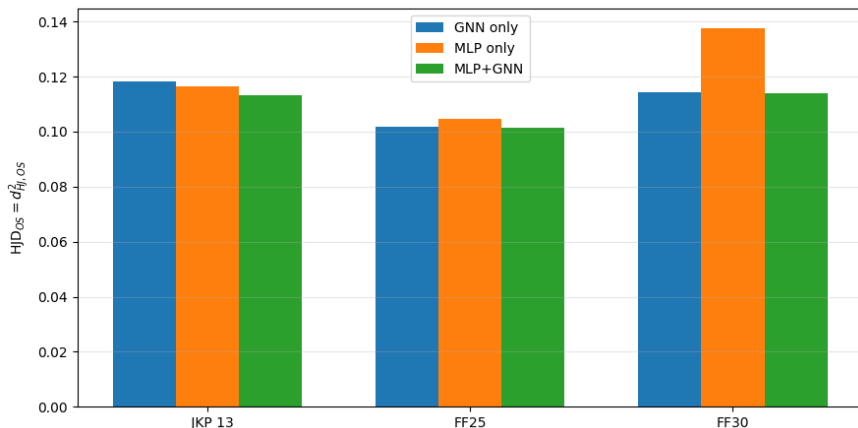


Each panel shows the distribution of absolute t -statistics $|\hat{t}_p|$ from the time-series regression described in Section 4.2, controlling for the five Fama–French factors (Fama and French, 2015b). The supply chain factor F_t is the long-minus-short portfolio of the $d = 3$ TransformerGNN, constructed following Equation (23). We estimate the regressions and compute the t -statistics with the entire data period, ranging from January 2003 through January 2024. The vertical dashed line marks the $|t| = 3$ threshold recommended by Chen and Zimmermann (2022) for robust inference. *Left (FF25)*: 25 portfolios double-sorted on size (market equity) and book-to-market ratio (Fama and French, 1993). *Center (FF30)*: 30 industry portfolios classified by four-digit SIC codes (Fama and French, 1997). *Right (JKP13)*: 13 long-minus-short anomaly theme factors of Jensen et al. (2023), spanning investment, debt issuance, seasonality, momentum, profitability, quality, low-risk, value, accruals, size, short-term reversal, idiosyncratic volatility, and skewness.

FF30: Industry portfolios. Figure 4b reports the analogous results for the 30 Fama–French industry portfolios. Over 86% of industries (26 out of 30) load significantly on the supply chain factor, with the largest exposures in “Personal and Business Services” ($|t| = 37.4$), “Business Equipment” ($|t| = 29.6$), and “Wholesale” ($|t| = 26.8$)—all industries with dense upstream and downstream supply chain relationships. The weakest loadings appear in “Business Supplies and Shipping Containers” and “Coal,” industries with more concentrated and geographically segmented supply chains. This cross-industry pattern is consistent with the network transmission mechanism: industries with denser supply chain connections are more exposed to higher-order propagation effects that linear benchmarks cannot capture, and command larger risk premia for bearing that network-transmitted risk.

JKP13: Anomaly theme factors. Figure 4c reports results for the 13 Jensen et al. (2023) theme factors, which aggregate well-known cross-sectional anomalies into representative long-minus-short portfolios. Over 69% of themes (9 out of 13) load significantly on the supply chain factor, with the largest exposures in “Investment” ($|t| = 66.2$), “Debt Issuance” ($|t| = 57.5$), and “Seasonality” ($|t| = 37.4$). The first two connections are economically direct: investment cycles propagate along supplier–customer chains (Cohen and Frazzini, 2008), and debt capacity depends on downstream revenue stability (Agca et al., 2022). The strong Seasonality loading is consistent with inventory cycle propagation along production networks (Barrot and Sauvagnat, 2016; Ahir et al., 2018)—

Figure 5: Out-of-Sample Hansen–Jagannathan Distance by Model and Test Asset Set



This figure plots the out-of-sample Hansen–Jagannathan distance (HJD) for three model specifications across three test asset sets. We compute the HJD for the out-of-sample period ranging from January 2017 through January 2024. *Test assets:* (1) the 13 anomaly theme factors of Jensen et al. (2023) (JKP13), which aggregate major cross-sectional return anomalies into theme portfolios; (2) the 25 Fama–French portfolios double-sorted on size and book-to-market (Fama and French, 1993) (FF25); and (3) the 30 Fama–French industry portfolios (Fama and French, 1997) (FF30). *Models:* characteristics-only 3-layer neural network (NN3 MLP, orange bars in the figure above), supply chain graph neural network (GNN, $d = 3$ TransformerGNN, blue bars in the figure above), and their combination (GNN+NN3 MLP, green bars in the figure above). Lower HJD indicates smaller joint pricing errors across the test portfolios. Numerical values are in Tables A.4–A.6.

firms time capital expenditures to match predictable seasonal demand from downstream customers, generating correlated return patterns across supply chain pairs.

Hansen–Jagannathan Distance. The time-series regressions above test whether the supply chain factor loads significantly on individual test portfolios in the entire data period, which may suffer from dimensionality of the test assets as well as lack of account of their covariance structure. We now complement this portfolio-by-portfolio evidence with the joint measure Hansen–Jagannathan distance (HJD) (Hansen and Jagannathan, 1991) of pricing accuracy exclusively on the January 2017 to January 2024 out-of-sample period. The HJD measures the minimum distance between a candidate stochastic discount factor (SDF) and the set of valid SDFs consistent with observed asset prices; a lower value indicates smaller joint pricing errors across all test portfolios simultaneously.²⁰ We compare three model specifications: a characteristics-only NN3, a supply chain GNN, and the combined GNN+NN3. Tables A.4–A.6 report the out-of-sample HJD and its reduction relative to the characteristics-only NN3 baseline for each test asset set.

The HJD results reveal a clear and economically interpretable pattern across the three test asset sets, summarized in Figure 5. For the 25 Fama–French size-value portfolios (FF25), adding

²⁰Formally, we approximate a linear SDF $m_t(\mathbf{b}) = 1 - \mathbf{b}^\top \mathbf{z}_t$, where $\mathbf{z}_t$ stacks either the five Fama–French control factors alone or the augmented set that also includes F_t GNN-produced supply chain factors. We estimate $\mathbf{b}$ on the training sample (through December 2016) and the out-of-sample HJD is the quadratic form $\widehat{\text{HJD}}_{OS} = \hat{\mathbf{e}}_{OS}^\top \hat{\mathbf{B}}_{OS}^\dagger \hat{\mathbf{e}}_{OS}$, where $\hat{\mathbf{e}}_{OS} = T_{OS}^{-1} \sum_{t \in OS} \hat{m}_t \mathbf{r}_t$ is the mean out-of-sample pricing error, $\hat{\mathbf{B}}_{OS} = T_{OS}^{-1} \sum_{t \in OS} \mathbf{r}_t \mathbf{r}_t^\top$ is the return second-moment matrix, $(\cdot)^\dagger$ denotes the Moore–Penrose pseudoinverse, and $\mathbf{r}_t \in \mathbb{R}^{N_{\text{test}}}$ is the vector of test-asset excess returns with $N_{\text{test}} \in \{25, 30, 13\}$ for FF25, FF30, and JKP13, respectively.

the supply chain GNN factor to the characteristics-only NN3 reduces the HJD by 2.77%, and the combined model reduces it by 3.15%. The gain is largest for FF30 (17.07% reduction), where industry membership and supply chain network position are tightly correlated, amplifying the GNN’s advantage over characteristic-only models.

The one exception is the JKP13 anomaly theme factors, for which the GNN alone yields a slight HJD increase of 1.63% relative to the characteristics-only NN3, while the combined GNN+NN3 model still improves by 2.75%. The JKP13 themes are designed to capture return anomalies driven primarily by own-firm financial characteristics—momentum, quality, and profitability—that are driven by firm-level signals rather than network transmission. The GNN’s incremental advantage is therefore concentrated in test assets that directly reflect supply chain network risk: the FF25 size-value sorts, which capture cross-sectional variation associated with firms’ positions within the supply chain network, and the FF30 industry portfolios, which aggregate firms whose revenues and costs are linked through input-output relationships.

4.3 Forecast Horizon and Transaction Cost Sensitivity

Supply chain information may contain both transitory and persistent components of firm risk: temporary disruptions resolve within months, while structural network effects could sustain predictability at longer horizons. We investigate whether these components can be separated empirically by forecasting h -month-ahead returns for $h \in \{0, 1, 3, 6\}$, holding the information set fixed at $\mathcal{F}_t = \sigma(\mathbf{X}_t, SCG_t)$ to isolate horizon sensitivity from any look-ahead in conditioning variables. Throughout, $Y_{i,t+h}$ denotes the excess return of firm i in month $t + h$, and the h -step predicted return $\hat{Y}_{i,t+h}^{(h)}$ is formed solely from $\mathcal{F}_t$, with no post- t information used. The baseline case $h = 0$ recovers the one-step model.

A model relying primarily on transitory effects, such as temporary inventory imbalances or local supply chain shocks, should produce long-short portfolios whose Sharpe ratios collapse quickly as the horizon extends. A model capturing persistent structural supply chain exposures would retain substantial portfolio value at multi-month horizons. We discriminate between these by comparing Sharpe ratios of long-minus-short portfolios sorted on h -step predicted returns across $h \in \{0, 1, 3, 6\}$.

We now discuss the multi-step model specification. Let $\mathcal{F}_t$ denote the information set available by the end of month t , comprising firm characteristics $\mathbf{X}_t$ and the observed supply chain graph SCG_t . The baseline model targets the contemporaneous conditional mean:

$$\hat{Y}_{it} = \hat{g}_i(\mathbf{X}_t, SCG_t),$$

where $\hat{g}_i(\cdot)$ is the trained GNN mapping and uses only variables in $\mathcal{F}_t$. For the h -step-ahead extension:

$$\hat{Y}_{i,t+h}^{(h)} = \hat{g}_i^{(h)}(\mathbf{X}_t, SCG_t),$$

where $\hat{g}_i^{(h)}$ is a model applied directly at horizon h using only information in $\mathcal{F}_t$, without post- t inputs.

Table 6: Long-minus-short portfolio return properties by forecast horizon, before transaction costs, using the $d = 3$ TransformerGNN.

Forecast horizon	ER	Volatility	Sharpe ratio
Baseline ($h = 0$)	0.1106	0.0047	1.3923
$h = 1$ month	0.0096	0.0037	0.1214
$h = 3$ months	0.0068	0.0032	0.0921
$h = 6$ months	0.0066	0.0028	0.0895

This table reports pre-trading-cost performances of long-minus-short portfolios sorted on multi-step predicted returns from the $d = 3$ TransformerGNN, constructed following Equation (23) (§4.1), with the sorting key replaced by $\hat{Y}_{i,t+h}^{(h)}$ for each horizon $h \in \{0, 1, 3, 6\}$ months. Columns report the portfolio mean return (%/month), return volatility (%/month), and Sharpe ratio over the out-of-sample period $\mathcal{D}_{\text{test}}$. $h = 0$ is the baseline one-step model. We report these quantities for the out-of-sample period ranging from January 2017 through January 2024.

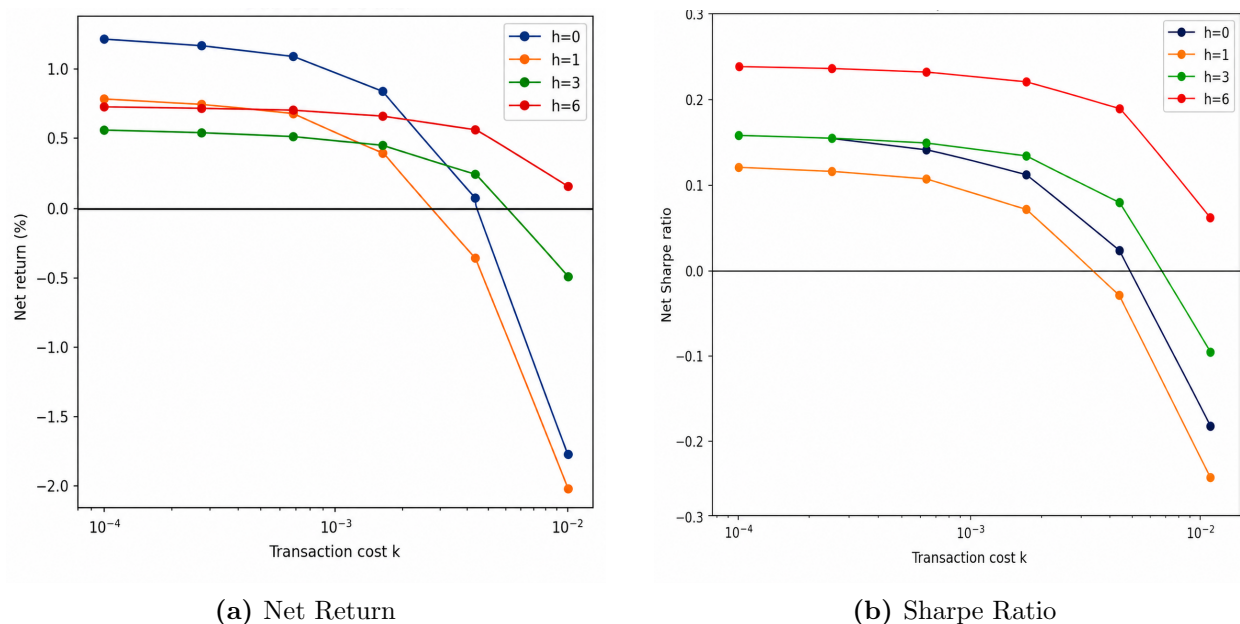
Table 6 reports long-minus-short portfolio properties before accounting for transaction costs. The Sharpe ratio collapses from 1.39 at the one-step baseline to 0.12 at $h = 1$ month, which is a 91% decline, and attenuates only marginally to 0.09 at $h = 3$ and $h = 6$. This near-complete deterioration within a single month’s extension establishes that the GNN’s return predictability is concentrated in contemporaneous cross-sectional pricing: the supply chain information in $\mathcal{F}_t$ that is priced by the market has largely been absorbed within one month, consistent with gradual diffusion of supply chain news documented in Cohen and Frazzini (2008). Portfolio return volatility declines monotonically with holding horizon, from 0.47%/month at $h = 0$ to 0.28%/month at $h = 6$ as reported in Table 6, because less frequent rebalancing averages out idiosyncratic short-term return fluctuations. The Sharpe ratio column reports the risk-adjusted returns, showing that $h = 0$ baseline still comes out as the pre-trading-cost winner since its mean return more than compensates for its higher volatility.

The volatility reduction at longer horizons in Table 6 should not be interpreted as evidence of superior signal stability. Holding portfolio weights fixed for h months between rebalances mechanically reduces month-to-month return variability through position staleness, independently of any genuine improvement in signal quality. For instance, a monthly strategy ($h = 0$) incurs this cost twelve times per year vs a semi-annual strategy ($h = 6$) only twice. To dissect the trade-off between high profitability for small h , reported in Table 6, and fewer portfolio rebalancings for longer h , we need to investigate how transaction costs interact with rebalancing frequency.

We find that aggregate turnover, defined as absolute value of differences between two adjacent weight vectors of stock holdings in rebalancing events (i.e. ℓ_1 distance between two weight vectors), is approximately 3.1–3.3 regardless of holding horizon (Appendix Table A.9). Thus, the key driver of aggregate cost differences across horizons would probably be the timing of rebalancing events, not the magnitude of per-event turnover.

For a fractional trading cost parameter κ , the effective annualized cost drag is approximately $\kappa \times \text{turnover} \times (12/h)$, which falls steeply as h increases even though per-event turnover is similar across horizons. A shorter holding horizon therefore triggers more rebalancing events per year

Figure 6: Net Return and Sharpe Ratio by Transaction Cost Rate and Holding Horizon.



In this figure, each panel plots a performance metric against the proportional transaction cost rate κ (x-axis, log scale) for each holding horizon $h \in \{0, 1, 3, 6\}$ months. *Left (Figure 6a):* net monthly-equivalent return (%). The $h = 0$ curve starts highest at low κ but declines more steeply with cost than the longer-horizon curves, with the $h = 6$ strategy overtaking it once transaction costs become sufficiently large. *Right (Figure 6b):* monthly-equivalent Sharpe ratio. In both panels, the statistics are computed from the test sample under the (κ, h) block-rebalancing strategy. We report these quantities for the out-of-sample period ranging from January 2017 through January 2024. Full numerical results by (κ, h) cell are reported in Appendix Tables A.7–A.8.

but not necessarily larger per-event turnover, since the portfolio weights at each rebalance reflect the same characteristic and graph inputs regardless of frequency. Whether frequent rebalancing is beneficial thus depends on the balance between the timeliness of the supply chain predictability and the aggregate transaction cost drag from more frequent rebalancing.

Therefore, we next document the portfolio returns and Sharpe ratios, net of trading costs, following Jensen et al. (2024), and trace how this asymmetry reshapes performance rankings as κ rises. Figure 6a plots monthly-equivalent net return against κ for each horizon. At $\kappa = 0$, the monthly strategy ($h = 0$) earns the highest net return (1.14%), reflecting its frequent exploitation of the contemporaneous supply chain predictability as evident in Table 6. Its cost sensitivity is however severe: $h = 0$ turns over approximately 39.3 times per year, so rising κ erodes its net return far more rapidly than longer horizons. The $h = 6$ strategy, accumulating only 6.6 rebalances per year, declines gradually. A crossing occurs at approximately $\kappa \approx 0.002$ (20 basis points): beyond this threshold, the 6-month strategy delivers higher net returns than the monthly strategy. At $\kappa = 0.01$ (100 basis points), $h = 0$ loses -2.14% per year while $h = 6$ retains $+0.18\%$.

Figure 6b reveals that the $h = 6$ strategy delivers the highest Sharpe ratio at *every* cost level κ , including $\kappa = 0$ where no transaction costs are paid. At $\kappa = 0$, the 6-month strategy achieves a Sharpe ratio of 0.241 versus 0.156 for $h = 0$ —a 54% advantage even in a frictionless

market. This is because the Sharpe ratio adjusts for return volatility, and less frequent rebalancing smooths idiosyncratic month-to-month fluctuations in the supply chain signal: the monthly strategy continuously rotates into new positions, each introducing short-term noise, while the 6-month strategy holds positions long enough to average over transitory variation. As shown in Table 6, portfolio volatility declines monotonically from 0.47%/month at $h = 0$ to 0.28%/month at $h = 6$ —a 40% reduction that more than offsets the decline in mean return on a risk-adjusted basis.

The Sharpe ratio advantage of $h = 6$ widens further as κ rises: at $\kappa = 0.01$, the monthly strategy produces a Sharpe ratio of -0.278 while the 6-month strategy remains positive at 0.060 . The monthly strategy maximizes gross returns only in a near-frictionless environment and is never the risk-adjusted optimum. For an institutional investor facing proportional costs above 20 basis points, $h = 6$ dominates on both net return and risk-adjusted performance. This horizon–cost complementarity supports the interpretation that supply chain return predictability contains a durable cross-sectional component that remains exploitable even when trading is infrequent.

5 Understanding GNN Representations of Supply Chains

A central question is whether GNN-based asset pricing models incorporate economically meaningful information about the structure of the supply chain network. While our discussion so far has focused on the GNN’s performance as a conditional mean estimator of returns, it is equally important to interpret the learned representations in order to understand the economic source of any gains from supply chain data. To this end, we examine the relationship between GNN-derived firm embeddings, directly tied to the model’s conditional mean excess return estimates, and both classical firm characteristics and network topology measures. This analysis assesses whether the models capture higher-order supply chain structure beyond what is contained in traditional cross-sectional characteristics alone.

5.1 Oversmoothing and Optimal Depth Selection in the Supply Chain Graph

As depth d grows, the *oversmoothing problem* (Chen et al., 2020) becomes the binding constraint on GNN performance: repeated message passing forces firm embeddings toward a common limit, independent of each firm’s characteristics or structural position in the supply chain network. This is qualitatively distinct from the bias–variance trade-off that governs depth selection in standard feedforward networks, where additional layers expand representational capacity at the cost of higher estimation variance. In GNNs operating on graphs with cyclic structure, additional layers do not expand capacity. Rather, they contract it by erasing the cross-sectional dispersion in embeddings that the model needs to distinguish firms with different risk exposures.

When embeddings converge, the GNN can no longer distinguish firms with different risk exposures, which directly limits its capacity to explain the cross-section of expected returns. In this subsection, we formally characterize the oversmoothing problem and demonstrate empirically how it manifests in our data and governs the selection of optimal network depth.

The mechanism can be understood through the cycle structure of the supply chain graph. When a firm lies on a cycle of length k , that is, it can reach itself in k hops along customer–supplier links, message passing at depths $d \geq k$ begins re-aggregating information from nodes already visited in earlier layers. New layers therefore repeatedly incorporate information that has already been absorbed into the firm’s embedding rather than introducing genuinely new signals. For example, a firm that is three hops away from itself through a cycle will have its embedding influenced by the same set of firms at layers 3, 6, 9, and so on, as the message-passing operation revisits the same neighborhood. Similarly, when two firms are connected through multiple paths in the network, their embeddings will increasingly reflect the same aggregated information as depth grows. Adding layers beyond this point therefore increases model parameters without expanding effective representational capacity: embeddings converge asymptotically toward a common vector, erasing the cross-sectional dispersion that the GNN needs to separate firms with different risk profiles and supply chain positions.

5.1.1 Measurement of Oversmoothing

Recall that we use $\mathbf{e}_{it}^{(l)} \in \mathbb{R}^q$ to denote the embedding of firm i at time t and layer l in our TransformerGNN. These embeddings encode each firm’s expected returns, incorporating both its own characteristics and the nonlinear aggregation of information from supply chain partners up to k hops away.

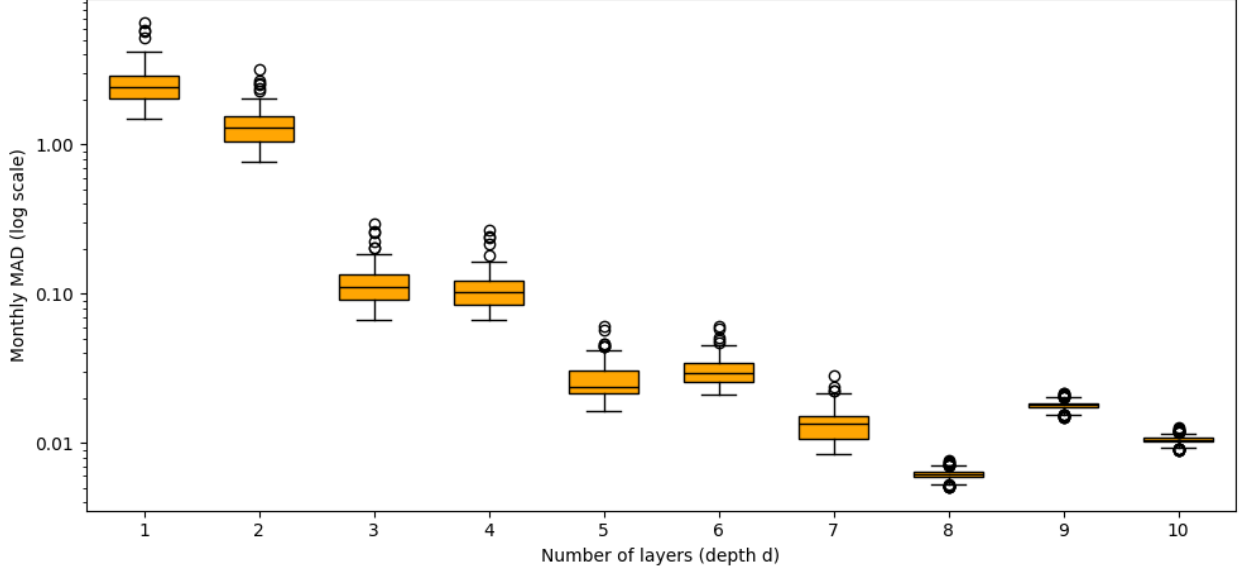
As the depth d increases, the GNN learns increasingly homogeneous embeddings for different nodes. This corresponds to capturing increasingly higher-order supply chain effects: layer 1 incorporates direct suppliers and customers, layer 2 adds suppliers-of-suppliers and customers-of-customers, and so on. If we increase to an arbitrarily high number of layers, then the traversal of suppliers may circle back to the starting node due to cycles in the supply chain graph.

To measure oversmoothing, we adopt the *mean pairwise distance* (MAD) introduced by [Chen et al. \(2020\)](#), due to its intuitive interpretation and direct relevance to our asset pricing application. For a cross-section of N_t firm embeddings $\{\mathbf{e}_{it}^{(d)}\}_{i=1}^{N_t}$ at depth d , MAD is the average pairwise Euclidean distance:

$$\text{MAD}^{(d)} = \frac{2}{N_t(N_t - 1)} \sum_{i < j} \|\mathbf{e}_{it}^{(d)} - \mathbf{e}_{jt}^{(d)}\|_2, \quad (24)$$

where $\|\cdot\|_2$ denotes the Euclidean norm. Smaller values of MAD indicate greater smoothing, with MAD approaching zero in the limit of complete oversmoothing where all firm embeddings become identical. This metric directly captures the loss of firm-specific heterogeneity that is relevant for asset pricing applications. In economic terms, MAD measures the dispersion of risk characterizations across firms: a high MAD indicates that the model distinguishes between firms with different risk exposures, while a low MAD suggests that the model assigns similar risk profiles to all firms, which can limit its effectiveness for explaining cross-sectional differences in expected returns. The choice of MAD as our primary oversmoothing metric is motivated by its clear economic interpretation: it quantifies the extent to which the GNN preserves the heterogeneity in firm risk exposures that is

Figure 7: Distribution of Cross-Sectional MAD vs. GNN Depth



This figure contains boxplots displaying the distribution of the cross-sectional monthly MAD for each depth. We compute the within-month MAD using all firms within a given month, and the box plots show the distributions of the MADs over all months. The y -axis is on a logarithmic scale to accommodate the wide range of MAD values across different d 's depths. We compute the distributions using the test embeddings which cover the out-of-sample period ranging from January 2017 through January 2024.

essential for generating meaningful return predictions.

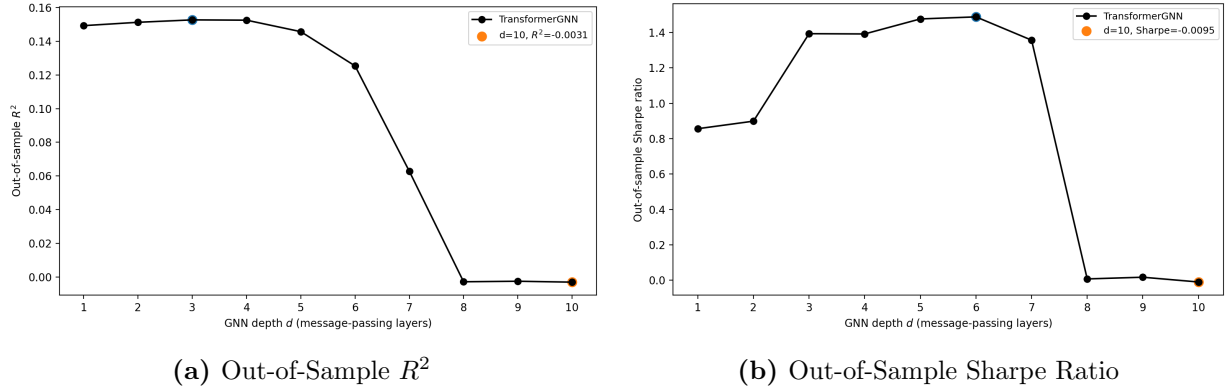
5.1.2 Empirical Evidence of Oversmoothing

From the empirical asset pricing perspective, our goal is to better understand the underlying structural reason behind our earlier results presented in Table 2 and Table 5, where the out-of-sample performance of GNN-managed portfolios peaks at $d = 3$ and $d \geq 8$ GNN models drastically worsened. Correspondingly, our analysis uses only the out-of-sample period to compute the MAD metric, connect it with R^2 , and finally to asset pricing outcomes of Sharpe ratios. This provides a comprehensive assessment of the oversmoothing phenomenon and its economic implications for return prediction.

We first examine how embedding dispersion varies with depth by computing the mean pairwise distance ($MAD^{(d)}$) of the final-layer embeddings for each depth d . These embeddings directly inform the conditional mean return estimates, and encode each firm's risk characterization, capturing how expected returns depend on the firm's position in the supply chain network and the nonlinear aggregation of information from connected firms.

Figure 7 reports the distribution of MAD values across depths, whereas Table A.10 reports the detailed statistics. The overall decline in MAD is dramatic: it falls from 2.61 at $d = 1$ to below 0.02 for all $d \geq 8$ —a greater than 99% reduction. More consequential for asset pricing is that the

Figure 8: Out-of-Sample R^2 and Sharpe Ratio vs. GNN Depth



Each panel plots a performance metric against GNN depth $d \in \{1, \dots, 10\}$. *Left (Figure 8a):* out-of-sample R^2 . The inverted-U shape peaks at $d = 3$ and falls to effectively 0 at $d \geq 8$, directly mirroring the MAD collapse in Table A.10. *Right (Figure 8b):* out-of-sample Sharpe ratio of the decile long-minus-short portfolio. The Sharpe ratio peaks at $d = 3$ and collapses at $d \geq 8$. We compute these quantities using data from the period ranging from January 2017 through January 2024. Both panels confirm that once oversmoothing erases embedding heterogeneity, the model loses its ability to explain the cross-section of returns or generate risk-adjusted trading profits.

collapse is essentially complete by $d = 8$: the mean and median MAD values of $d = 8, 9, 10$ are much closer to zero relative to the scale of variation at $d = 3$ ($\text{MAD}^{(3)} = 0.120$). At these depths, the GNN assigns embeddings that are nearly identical across firms regardless of their characteristics or network position—the model has lost virtually all capacity to discriminate one firm’s expected return from another’s. Figure 7 shows that the interquartile range of monthly MAD values, already narrow at $d = 5$, collapses to a tight band near zero for $d \geq 8$. This MAD collapse is the root cause of the near-zero out-of-sample R^2 and Sharpe ratios at deep networks, as we show directly in the two subsections that follow.

The declining trend in MAD as depth increases is broadly consistent with oversmoothing theory, although the ordering is not strictly monotone at deep layers ($d \geq 5$), where non-monotone reversals arise from cycles in local subgraphs. As depth increases, firms connected through multiple paths receive increasingly overlapping signals: two firms three hops from a common supplier receive nearly identical aggregated information once $d > 3$. The $d = 3$ specification captures up to third-order supply chain linkages, namely the point at which incremental higher-order information is still available before redundant cycle traversal begins to dominate.

Out-of-Sample R^2 and Oversmoothing. We next examine how the loss of embedding dispersion affects predictive accuracy for firm-level returns. Figure 8a plots out-of-sample R^2 against GNN depth d , computed using the rolling-window evaluation framework of Section 3.2.

Predictive accuracy follows an inverted-U in depth. R^2 rises from $\approx 14.9\%$ at $d = 1$ to $\approx 15.3\%$ at $d = 3$ as additional layers capture higher-order supply chain relationships. Beyond $d = 3$, R^2 gradually declines and eventually falls sharply to near-zero at $d \geq 8$. This collapse coincides precisely with the depth range at which Table A.10 shows MAD falling below 0.02:

once oversmoothing renders firm embeddings nearly identical, the model retains no firm-specific information to price, and R^2 reaches its floor. The boxplot in Figure 7 makes the timing of the collapse visible month by month—the distribution of MAD tightens to near-zero at the same depths where R^2 reaches its floor.

Out-of-Sample Sharpe Ratio and Oversmoothing. Finally, we assess the economic significance of oversmoothing by examining its impact on the Sharpe ratio of the long-short decile portfolio strategy. Figure 8b presents the relationship between GNN depth and the out-of-sample Sharpe ratio. The x-axis shows the number of layers d , while the y-axis reports the Sharpe ratio computed from the decile portfolio formed by sorting firms on GNN-predicted returns. This metric captures the economic value of the model’s predictions, measuring the risk-adjusted returns achievable by trading on the model’s predicted returns.²¹

The Sharpe ratio confirms the economic cost of oversmoothing. It rises from $d = 1$, peaks at $d = 3$, then declines monotonically, collapsing to near zero at $d \geq 8$ —the same depths at which Table A.10 shows MAD falling below 0.02. Once embedding heterogeneity is lost, the decile-sorting strategy loses all ability to discriminate firms by expected return and the long-short spread disappears. In other words, we expect to find low MAD at $d \geq 8$ not as a statistical artifact, but rather as an economically sensible product of supply chain graphs being interconnected. This connectedness then causes near-zero R^2 and Sharpe ratios at $d \geq 8$ depths, tracing the performance collapse to the topology of the supply chain graph rather than to estimation noise or overfitting.

5.1.3 TransformerGNN and Oversmoothing

Our TransformerGNN architecture, which uses multi-head attention mechanisms rather than simple graph convolutions, provides some mitigation of the oversmoothing problem. The attention mechanism in Equation (27) allows each firm to dynamically weight the importance of different supply chain partners, potentially preserving more distinctive information than uniform message passing. This is economically meaningful because it allows the model to learn that certain supply chain relationships are more important for expected returns than others—for example, a firm may be more sensitive to shocks from its largest customer than from its smallest supplier. However, Chen et al. (2020) demonstrate that even attention-based GNNs are not immune to oversmoothing, as the aggregation operation in Equation (28) still combines information across increasingly large neighborhoods, and the attention weights themselves may converge to uniform distributions as depth increases.

The empirical evidence confirms that the attention mechanism delays but does not prevent oversmoothing. From $d = 3$ to $d = 10$, MAD declines by 91.3%, while the out-of-sample Sharpe ratio falls from 1.3923 to -0.0095 and R^2 from 0.1527 to -0.0030 . Beyond $d = 3$, additional layers aggregate redundant signals from increasingly distant supply chain partners, and the attention

²¹Throughout this subsection, “predicted returns” refers to the model’s estimated conditional mean of stock i ’s excess return in period t , formed from supply-chain and characteristic data available at the portfolio formation date; no future information is used at any stage of the out-of-sample evaluation.

weights converge toward uniform distributions—reducing the node-specific weighting that attention provides at shallower depth. The $d = 3$ specification lies at the efficient frontier where higher-order supply chain transmission is captured while firm-specific heterogeneity is preserved.

5.2 Graph and firm characteristics learned by GNN embeddings

To gain insight into what the GNN models have learned, we analyze the firm-level embeddings produced by the network. Recall that when we introduced conditional mean estimation in §2.3, the final embedding is $\mathbf{e}_{it}^{(d)} = h_{it}^{(d)} \in \mathbb{R}^q$, obtained after d rounds of message passing and aggregation. This embedding is the last hidden-layer representation, summarizing both a firm’s own characteristics and the information aggregated from supply chain neighbors across all d layers. The conditional mean excess return is then obtained by applying a linear transformation to this final embedding, as shown in equation (7).

These embeddings are linearly related to the model’s estimates of the conditional mean excess returns and thus summarize the information—both from characteristic-managed portfolio returns and the supply chain network—that the model deems relevant for pricing. We investigate whether these embeddings encode systematic economic variation by testing for differences in traditional firm characteristics (e.g., book-to-market ratio) and network structural measures (e.g., degree centrality).

The link between embeddings and predictions is direct: the GNN’s forecast of firm i ’s conditional mean excess return at time t is obtained by applying a linear layer to $\mathbf{e}_{it}^{(d)}$ in equation (7). More technical details are included in §B.1. Thus, the embedding $\mathbf{e}_{it}^{(d)}$ contains all the information the model relies on, and the prediction is simply a weighted linear combination of its coordinates. This makes the embeddings natural objects for interpretation: by studying their statistical variation across firms, we can infer what economic and structural information the GNN has learned and encoded.

Goal of the ANOVA F-Test. We aim to test whether the GNN can distinguish among firms based on variation in specific economic variables. The embeddings produced by the GNN summarize the information extracted by the nonlinear Transformer model, and our goal is to assess whether they contain meaningful economic content. In particular, we evaluate whether firms grouped by their embeddings exhibit systematic differences in network structures—such as degree centrality—or in firm characteristics identified by Freyberger et al. (2020) as relevant for expected returns. Demonstrating such differences would link the GNN’s learned representations to interpretable dimensions of firm heterogeneity.

K-Means Grouping of the Firms. We apply K-means clustering (MacQueen, 1967) to the principal components of the learned embeddings at each time period, with 2 groups. The clustering is performed by solving:

$$\{\mathcal{G}_1, \mathcal{G}_2\} = \arg \min_{\mathcal{G}_1, \mathcal{G}_2} \sum_{k=1}^2 \sum_{i \in \mathcal{G}_k} \|\mathbf{e}_{it}^{(d)} - \bar{\mathbf{e}}_{k,t}\|_2^2,$$

where $\mathbf{e}_{it}^{(d)}$ is the embedding of firm i at time t , and $\bar{\mathbf{e}}_{k,t}$ is the centroid of cluster k . We begin with $K = 2$ because, if significant differences in characteristics are already evident between just two clusters, this minimal grouping provides strong evidence of economic relevance. Higher values of K can then be considered as an extension.

Null Hypothesis and F-Test Setup. For a given characteristic x_{it} , let $\bar{x}_{1t}$ and $\bar{x}_{2t}$ denote the cross-sectional means for clusters $\mathcal{G}_1$ and $\mathcal{G}_2$ at time t . We test:

$$H_0 : \bar{x}_{1t} = \bar{x}_{2t} \quad \text{vs.} \quad H_A : \bar{x}_{1t} \neq \bar{x}_{2t}.$$

The F-statistic is given by:

$$\text{F-stat} = \frac{\sum_{k=1}^2 n_k (\bar{x}_{k,t} - \bar{x}_t)^2 / (K - 1)}{\sum_{k=1}^2 \sum_{i \in \mathcal{G}_k} (x_{it} - \bar{x}_{k,t})^2 / (N - K)},$$

where n_k is the number of firms in cluster k , $\bar{x}_t$ is the overall mean at time t , and $K = 2$. Under the null, F-stat follows an F distribution with $(K - 1, N - K)$ degrees of freedom.

We first examine graph-structural characteristics. Degrees are defined as follows: total degree equals the sum of a firm’s in-degree (number of distinct suppliers) and out-degree (number of distinct customers) in the directed supply chain graph. Table 7 reports the corresponding F-statistics and p-values. All degree measures exhibit extremely high F-statistics and p-values below 0.0001, rejecting the null hypothesis. This indicates that the GNN embeddings capture important structural information in the supply chain network—differences in degree centrality are reflected in the learned representations, and these differences are relevant for conditional mean return estimation.

Table 7: F-statistics for network degree characteristics across $K = 2$ embedding clusters.

Characteristic	F-stat	p-val
Avg. Total Degree	449.037	< .0001
Avg. Out-Degree	448.895	< .0001
Avg. In-Degree	387.518	< .0001

This table reports the F-statistics and p-values of the three graph structure characteristics across firms that live in the two embedding clusters. We compute the degree characteristics using the embeddings which cover the entire period ranging from January 2003 through January 2024.

Many firm-level characteristics—most notably cash flow measures (Free CF, [Hou et al. \(2020\)](#)) and the size-to-market ratio (LME, [Fama and French \(1993\)](#)) and Tobin’s Q (Q, [Kaldor \(1966\)](#))—also exhibit high F-statistics, the top 20 of which are shown in Figure 9. These variables are defined consistently with the asset pricing literature, and their significance suggests that the GNN embeddings encode dimensions of firm fundamentals that extend beyond purely network-based information. The full list of variables and definitions is provided in our [supplemental documentation](#).

Figure 9: Top 20 most significant characteristics by F-statistic across $K = 2$ embedding clusters.

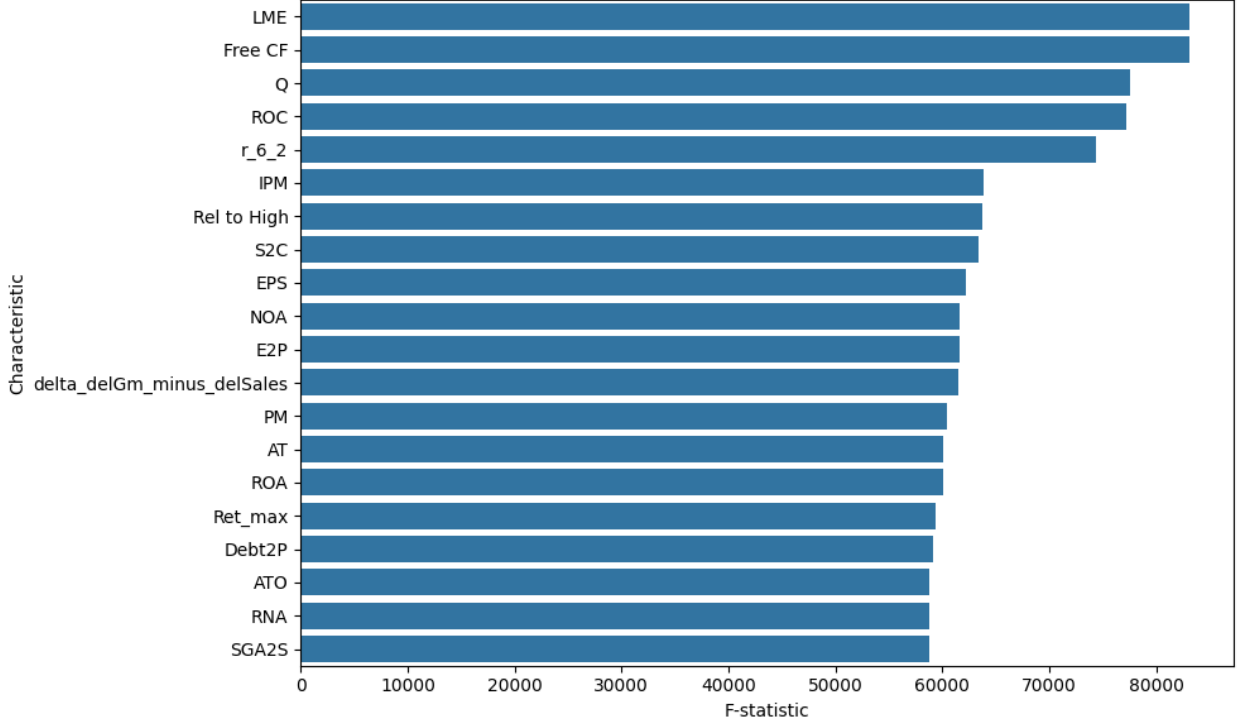


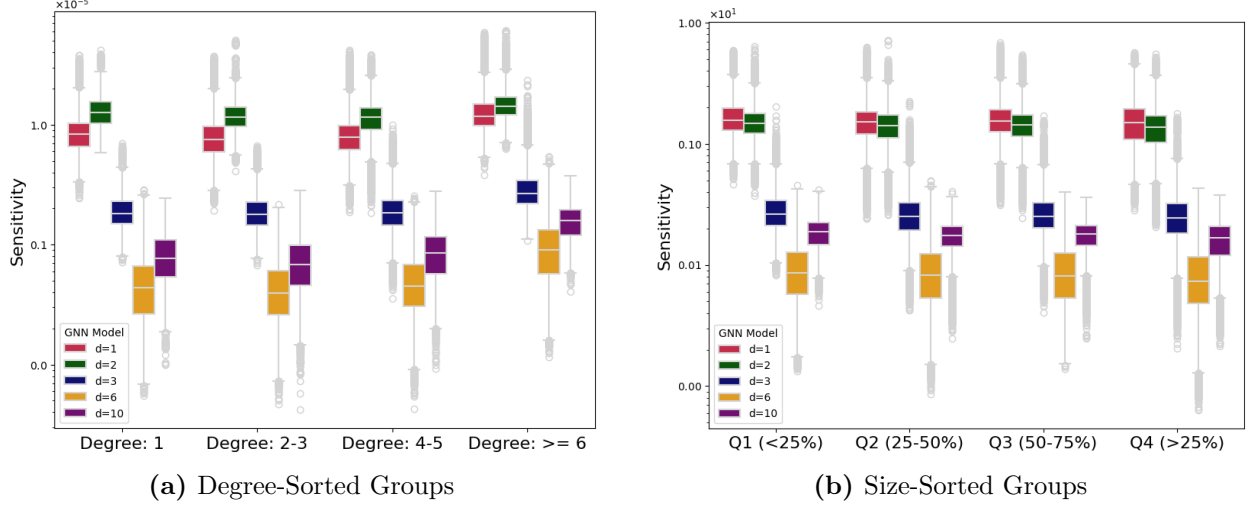
Figure of the 20 highest F-statistics of the firm characteristics across the two clustered embeddings from the trained GNN models. We compute the degree characteristics using the embeddings which cover the entire period ranging from January 2003 through January 2024. Note that all these F-statistics are significantly non-zero under the 0.0001 Type I error significance threshold. The names of these variables are available [online](#).

5.3 Asset Pricing Sensitivity to Perturbed Supply Chain Graph

Supply chain data is inherently incomplete: FactSet captures only a fraction of actual buyer-supplier relationships, and the coverage of any given firm fluctuates as new filings disclose previously unrecorded links. A key question is whether our GNN predicted returns inherit this measurement error—whether missing edges in the supply chain graph translate into material shifts in estimated pricing signals. We address this with a Monte Carlo edge-deletion experiment across two firm partitions: (i) *degree centrality*, testing whether sensitivity concentrates in structurally central firms; and (ii) *firm size*, testing whether the pricing effect is concentrated in large-cap supply chain relationships. The two partitions have distinct economic content: the former identifies which firms’ supply chain relationships drive the model’s predicted returns; the latter tests whether the results are driven by large-cap firms rather than by supply chain network structure per se.

We partition firms into four groups under each scheme—by network degree ($\{1\}$, $\{2, 3\}$, $\{4, 5\}$, $\{\geq 6\}$) and by rolling market-capitalization quartile (Small-cap, Mid-small Cap, Mid-large Cap, Large-cap, from smallest to largest market equity at time t)—and sample 100 firms per group in each

Figure 10: Sensitivity Analysis of Monte Carlo Experiments on SCG: Degree- and Size-Sorted Groups



Each panel shows the distribution of sensitivity $\tilde{S}_{it}$ (defined in Equation 25) across $B = 100$ Monte Carlo edge-deletion simulations using all data ranging from January 2003 through January 2024. The y -axis is on a log scale. *Left (Figure 10a)*: firms sorted into four degree groups by supply chain degree centrality. The x -axis shows degree groups with increasing centrality from left to right. *Right (Figure 10b)*: firms sorted into four rolling market-capitalization groups (Small-cap, Mid-small Cap, Mid-large Cap, Large-cap) from smallest to largest market equity at time t .

simulation draw.²² In each iteration, we delete one edge connected to a randomly chosen node within the selected group and measure the absolute change in the GNN predicted return:

The sensitivity of the predicted return for firm i to the b th edge deletion at time t is:

$$\tilde{S}_{it}^{(b)} = \left| \hat{g}_i(\mathbf{X}_t, SCG_t^{(b)}) - \hat{g}_i(\mathbf{X}_t, SCG_t) \right|, \quad (25)$$

where $SCG_t^{(b)}$ is the b th perturbed graph, $b = 1, \dots, B$. We repeat each experiment $B = 100$ times per group to marginalize over specific edge and node choices; the resulting distribution of $\tilde{S}_{it}^{(b)}$ captures both the average level and the variability of sensitivity within each subgroup. Figure 10 reports the full distributions as box plots (degree-sorted in Figure 10a, size-sorted in Figure 10b); Table 8 reports the corresponding means and standard deviations.

The degree-sorted results show that supply chain-central firms are associated with substantially larger sensitivity to missing links. For the $d = 3$ model, mean sensitivity for high-degree hubs (degree ≥ 6) is 55% higher and dispersion is 114% greater than for the least-connected firms (Panel (a) of Table 8). The widening box plots in Figure 10a confirm that high-degree nodes generate not only larger but also more variable responses to edge deletion, and the pattern holds across all model depths.

²²More details on the evolution of firm counts per degree group are reported in Figure A.2. Degree-group counts are unequal and time-varying; fixing the sample at 100 nodes per group per draw eliminates this confound and ensures comparability across groups of unequal size.

Table 8: Mean and Standard Deviation of Monte Carlo Sensitivity Experiments on SCG

<i>Panel (a): Degree-sorted groups</i>				
Model	Degree: 1	Degree: 2–3	Degree: 4–5	Degree: ≥ 6
TransformerGNN ($d = 1$)	0.88 (0.34)	0.83 (0.37)	0.86 (0.38)	1.32 (0.58)
TransformerGNN ($d = 2$)	1.33 (0.41)	1.23 (0.38)	1.17 (0.41)	1.52 (0.51)
TransformerGNN ($d = 3$)	0.20 (0.07)	0.20 (0.07)	0.20 (0.09)	0.31 (0.15)
TransformerGNN ($d = 6$)	0.05 (0.03)	0.05 (0.03)	0.05 (0.03)	0.10 (0.06)
TransformerGNN ($d = 10$)	0.08 (0.04)	0.08 (0.04)	0.09 (0.04)	0.16 (0.05)
<i>Panel (b): Size-sorted groups</i>				
Model	Small-cap	Mid-small Cap	Mid-large Cap	Large-cap
TransformerGNN ($d = 1$)	1.71 (0.64)	1.59 (0.68)	1.65 (0.64)	1.60 (0.73)
TransformerGNN ($d = 2$)	1.57 (0.52)	1.47 (0.57)	1.48 (0.51)	1.42 (0.62)
TransformerGNN ($d = 3$)	0.29 (0.13)	0.28 (0.15)	0.28 (0.13)	0.27 (0.16)
TransformerGNN ($d = 6$)	0.10 (0.06)	0.10 (0.06)	0.10 (0.05)	0.09 (0.06)
TransformerGNN ($d = 10$)	0.19 (0.05)	0.17 (0.05)	0.18 (0.05)	0.16 (0.06)

Reported values are mean sensitivity and, in parentheses, standard deviation, in units of 10^{-5} . Sensitivity is defined in (25); results are from $B = 100$ Monte Carlo edge-deletion simulations using all data from January 2003 through January 2024. *Panel (a)*: columns show four supply chain degree groups with increasing centrality from left to right. *Panel (b)*: columns show four rolling market-capitalization quartiles at each date t , with Size Group 1 (Size Group 4) comprising the smallest (largest) firms by market equity; 100 firms are sampled per group per draw. The absolute scale of $\tilde{S}$ reflects the specific perturbation protocol and should not be interpreted in isolation; the relevant comparisons are across groups within a given model (across columns) and across depths for a given group (across rows).

A plausible mechanism is network propagation: deleting an edge of a high-degree hub disrupts the message-passing path of every firm within d hops, whereas the same perturbation for a peripheral firm remains contained. Model depth governs how far this propagation reaches. Shallow models are sensitive even for low-degree firms: the $d = 1$ and $d = 2$ specifications produce sensitivity levels four to seven times higher than $d = 3$ for the least-connected nodes (Table 8). The $d = 3$ specification reduces mean sensitivity by 83% relative to $d = 2$ on average across all degree groups, with residual sensitivity concentrated in high-degree nodes. Deepening to $d = 6$ lowers sensitivity further, but at $d = 10$ the oversmoothing regime becomes prominent: mean sensitivity for high-degree firms rebounds by 60% relative to $d = 6$, consistent with over-averaged embeddings becoming more susceptible to the perturbations that remain after homogenization.

The size-sorted experiment yields a starkly different pattern. For the $d = 3$ model, mean sensitivity varies by only 7% across the four market-capitalization quartiles and standard deviations overlap throughout (Panel (b) of Table 8). The depth-sensitivity schedule—the 81% reduction from $d = 2$ to $d = 3$ and the $d = 10$ oversmoothing rebound—is replicated across all four size groups without material variation.

This uniformity across size groups reflects a distinction between network position and firm size. Market capitalization and graph degree need not move together: a mid-size manufacturer

with many direct counterparties may occupy a high-degree position in the SCG, while a large conglomerate maintaining only a handful of direct supply chain links has low graph degree. To the extent the GNN propagates information through graph topology rather than value-weighted aggregation, sensitivity patterns should track network structure more closely than firm size—which is precisely what we observe.

The two experiments collectively suggest that network position is more strongly associated with GNN sensitivity to supply chain data gaps than firm economic scale. High-degree firms exhibit 55% higher mean sensitivity and 114% greater dispersion relative to peripheral firms, while the cross-size spread in mean sensitivity is 7%—less than one-eighth of the cross-degree spread. This pattern is consistent with the GNN learning predominantly from graph topology rather than firm size, though the uniform single-edge-deletion design does not allow us to fully rule out alternative explanations based on correlated perturbations or systematic differences in edge importance across size groups.

6 Conclusion

We have introduced a graph neural network framework for systematic asset pricing with supply chains. By nonlinearly aggregating firm characteristics across d -hop production network neighborhoods, the framework captures higher-order supply chain transmission effects that own-firm and pairwise models cannot detect. The $d = 3$ specification achieves the optimal balance between bias reduction from higher-order linkages and the oversmoothing constraint imposed by the supply chain graph’s cyclic topology. After orthogonalizing against neural network, Fama–French, PCA, and linear supply chain benchmarks, we find that the residual GNN portfolio earns a Sharpe ratio of 1.07, confirming pricing capability incremental to all existing predictors.

As the first comprehensive asset pricing study to leverage a GNN, we highlight the distinct property of oversmoothing that drives the model towards homogeneity as the number of layers d becomes too large. Since the MAD flattens as $d \geq 8$, the GNN corresponding to those higher depths predicts similar returns for the cross-section, and the GNN-managed long-minus-short portfolio earns no excess returns. In addition, we have shown that sensitivity to missing supply chain links concentrates in high-centrality firms—mean sensitivity for high-degree hubs is 55% higher than for peripheral firms—while remaining uniform across the firm-size distribution. This finding identifies the position of a firm within the network as the primary driver of pricing sensitivity.

Although the focus of this paper has been on asset pricing, the framework applies broadly to economic problems with graph-panel (bimodal/*multimodal*) data²³: it jointly learns from the network structure and time-varying covariates and accommodates objectives expressible as weighted sums of moment conditions. The depth parameter d provides a data-driven rule for aggregating covariates within d -hop neighborhoods. In our case, we use a financial economics-informed objective to combine two modes of SCG data and firm-level returns and characteristics panel. Our framework

²³The terms *multimodal* data or *multimodal* learning stem from the ML community (Ngiam et al., 2011), where data sources such as videos, photos and texts are in different *modes* and not naturally compatible, so researchers need to propose intelligently constructed models to jointly learn from different *modes* of data.

generalizes to any problem of graph-panel bimodal data as long as the economic objective can be written as weighted sum of moment conditions. Economic problems involving graphs are ubiquitous (e.g., risk propagation in networks (Acemoglu et al., 2012) and causal inference under interference (Hudgens and Halloran, 2008)), and typically involve rich node- and time-varying covariates. By taking a step toward a conceptual framework for graph-panel multimodal learning, we believe our methodology is broadly applicable to problems involving graphs and panel data.

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Appendix

A Alternative graph representation

To understand why the graph is an efficient way to represent jointly supply chain and firm characteristics information, we consider a naive alternative: for each pair of firm (i, j) , we have returns and characteristics $(Y_{it}, \mathbf{x}_{it}, Y_{jt}, \mathbf{x}_{jt}) \in \mathbb{R}^{2(p+1)}$, while there are $|E_t|$ pairs. This adds up to a naive vector representation of all pairwise firm information as a D_t^{Naive} -dimensional vector of data:

$$D_t^{\text{Naive}} = 2(p+1) \times |E_t|. \quad (26)$$

Since each node in our graph has at least one edge by definition, $|E_t| > |V_t|$, and $D_t^{\text{Naive}} > D_t^{\text{Graph}}$ and the graph representation is more succinct. In the worst case, $|E_t|$ can be as high as $\binom{|V_t|}{2} = \frac{1}{2}|V_t|(|V_t|-1)$, so that the dimensional difference between D_t^{Naive} and D_t^{Graph} is on the order of $p \times |V_t|^2$.

Our next step is to leverage flexible ML methods to jointly extract information from $\mathbf{X}_t$ and SCG_t , aiming to explain the cross-section of average returns using our data representations.

B Appendix: Mathematical details of graph neural networks

B.1 Transformer Architecture

Recall from §2.3 that the embedding of firm i at time t is the final hidden-layer state $h_{it}^{(d)} \in \mathbb{R}^q$ produced after d rounds of message passing, as defined in Equations (4)–(5). This embedding encodes both firm-specific characteristics and supply chain network information, and the conditional mean return estimate is a linear function of this embedding (Equation (6)).

In our model, we enhance the GNN learning by incorporating a Transformer architecture proposed in Shi et al. (2021), which allows the GNN to more effectively capture complex relationships within the supply chain network. A multi-head Transformer-based approach introduces cross-attention mechanisms that assign different weights to each node’s neighboring messages, dynamically learning the importance of various connections.

For each firm i at time t and layer l with $l = 1, 2, \dots, d$, the H -multi-head Transformer mechanism computes a set of attention weights that determine the importance of each neighbor $j \in \mathcal{N}_t(i)$ in the update process. The Transformer-based computation for node i can be represented as follows:

- **Message with Multi-Head Attention** (f_{msg}): For a node i , the message from a neighboring node j is computed using multiple attention heads. Each head ℓ independently computes

attention weights and corresponding messages:

$$\mathbf{m}_{ijt}^{(l,\ell)} = \text{softmax} \left(\frac{(\mathbf{W}_Q^{(l,\ell)} \mathbf{h}_{it}^{(l)})^\top (\mathbf{W}_K^{(l,\ell)} \mathbf{h}_{jt}^{(l)})}{\sqrt{d_k}} \right) \mathbf{W}_V^{(l,\ell)} \mathbf{h}_{jt}^{(l)}, \quad (27)$$

where $\mathbf{W}_Q^{(l,\ell)}$, $\mathbf{W}_K^{(l,\ell)}$, and $\mathbf{W}_V^{(l,\ell)}$ are learnable query, key, and value matrices for head ℓ in layer l ; $\mathbf{h}_{it}^{(l)}$ and $\mathbf{h}_{jt}^{(l)}$ are the embeddings of firms i and j ; and d_k is the key dimension.

The softmax function normalizes the attention scores across all neighbors.

- **Aggregation Across Attention Heads (AGG):** The aggregated message for node i is computed by concatenating the outputs of all attention heads and applying a learnable linear transformation:

$$\mathbf{m}_{it}^{(l)} = \mathbf{W}_O^{(l)} \cdot \text{concat} \left(\mathbf{m}_{it}^{(l,1)}, \mathbf{m}_{it}^{(l,2)}, \dots, \mathbf{m}_{it}^{(l,H)} \right), \quad (28)$$

where H is the number of attention heads, $\mathbf{m}_{it}^{(l,\ell)} = \sum_{j \in \mathcal{N}_t(i)} \mathbf{m}_{ijt}^{(l,\ell)}$ is the aggregated message for head ℓ , and $\mathbf{W}_O^{(l)}$ is a learnable matrix that combines the outputs of all heads for the l th layer.

- **Update (f_{upd}):** The node state is updated using a feedforward neural network (FFN) with residual connections and normalization, as is typical in transformers:

$$\mathbf{h}_{it}^{(l+1)} = \text{LayerNorm} \left(\mathbf{h}_{it}^{(l)} + \text{FFN} \left(\text{LayerNorm}(\mathbf{h}_{it}^{(l)} + \mathbf{m}_{it}^{(l)}) \right) \right), \quad (29)$$

where the first residual connection combines the previous node state $\mathbf{h}_{it}^{(l)}$ with the aggregated message $\mathbf{m}_{it}^{(l)}$, and the second residual connection refines the updated state through a position-wise feedforward network. He et al. (2016) proposed the residual connection (i.e., the one adding $\mathbf{h}_{it}^{(l)}$) to help mitigate gradient vanishing issues, facilitating the learning of deeper networks, and it is well-recognized as a popular technique in ML.

Our choice of FFN leverages ELU activation function and the LayerNorm regularization (Bader et al., 2016) to compute the aggregated information at each node by applying a series of transformations. The FFN for node i at time t and layer l can be written as:

$$\text{FFN}(\mathbf{z}_{it}^{(l)}) = \mathbf{W}_2^{(l)} \cdot \text{ELU}(\mathbf{W}_1^{(l)} \mathbf{z}_{it}^{(l)} + \mathbf{b}_1^{(l)}) + \mathbf{b}_2^{(l)},$$

where:

- $\mathbf{z}_{it}^{(l)}$ is the input to the FFN for node i at time t , typically $\mathbf{z}_{it}^{(l)} = \text{LayerNorm}(\mathbf{h}_{it}^{(l)} + \mathbf{m}_{it}^{(l)})$, combining the previous node state $\mathbf{h}_{it}^{(l)}$ and the aggregated message $\mathbf{m}_{it}^{(l)}$.
- $\mathbf{W}_1^{(l)}, \mathbf{W}_2^{(l)}$ are learnable weight matrices in the FFN layers; and $\mathbf{b}_1^{(l)}, \mathbf{b}_2^{(l)}$ are learnable bias vectors.

- $\text{ELU}(x)$: “Exponential linear unit” is a nonlinear entry-wise activation function defined in [Clevert \(2015\)](#) of

$$\text{ELU}(x) = \begin{cases} x & x > 0 \\ \exp(x) - 1 & x \leq 0 \end{cases}. \quad (30)$$

C Appendix: Empirical Asset Pricing

C.1 PCA Benchmark Construction for Firm-Level Return Prediction

This subsection details the PCA benchmark used in Table 1, following the benchmark construction in [Giglio and Xiu \(2021\)](#). The procedure follows a transparent two-step design: first, extract latent components from characteristic-managed long-short portfolios; second, map those components into firm-level expected returns through estimated loadings.

Step 1: PCA on characteristic-managed top-minus-bottom portfolios. For each characteristic $c \in \{1, \dots, p\}$ and month t , form an equal-weighted decile long-short portfolio return

$$M_{ct}^{\text{LS}} \equiv R_{ct}^{\text{Top}} - R_{ct}^{\text{Bottom}}, \quad (31)$$

where R_{ct}^{Top} and R_{ct}^{Bottom} are returns on the top and bottom deciles sorted on characteristic c . Stack these into $\mathbf{M}_t^{\text{LS}} = (M_{1t}^{\text{LS}}, \dots, M_{pt}^{\text{LS}})^\top \in \mathbb{R}^p$.

Using only the training sample $t \in \mathcal{D}_{\text{train}}$, standardize each series:

$$\widetilde{\mathbf{M}}_t^{\text{LS}} = \mathbf{D}^{-1}(\mathbf{M}_t^{\text{LS}} - \bar{\mathbf{M}}^{\text{LS}}), \quad \bar{\mathbf{M}}^{\text{LS}} = \frac{1}{|\mathcal{D}_{\text{train}}|} \sum_{t \in \mathcal{D}_{\text{train}}} \mathbf{M}_t^{\text{LS}}, \quad (32)$$

where $\mathbf{D}$ is diagonal with training-sample standard deviations. Define

$$\widehat{\Sigma}_M = \frac{1}{|\mathcal{D}_{\text{train}}|} \sum_{t \in \mathcal{D}_{\text{train}}} \widetilde{\mathbf{M}}_t^{\text{LS}} \widetilde{\mathbf{M}}_t^{\text{LS}\top}. \quad (33)$$

Let $\widehat{\Sigma}_M = \widehat{\mathbf{V}} \widehat{\Lambda} \widehat{\mathbf{V}}^\top$ be the eigendecomposition, with eigenvalues sorted in descending order, and denote $\widehat{\mathbf{V}}_K = (\widehat{\mathbf{v}}_1, \dots, \widehat{\mathbf{v}}_K) \in \mathbb{R}^{p \times K}$. The K -dimensional PCA factor at month t is

$$\widehat{\mathbf{f}}_t = \widehat{\mathbf{V}}_K^\top \widetilde{\mathbf{M}}_t^{\text{LS}} \in \mathbb{R}^K. \quad (34)$$

Step 2: Firm-level return projection on PCA factors. For each firm i , estimate training-sample loadings by OLS:

$$(\hat{\alpha}_i, \hat{\beta}_i) = \arg \min_{\alpha, \beta} \sum_{t \in \mathcal{D}_{i, \text{train}}} \left(Y_{it} - \alpha - \beta^\top \widehat{\mathbf{f}}_t \right)^2, \quad (35)$$

where $\mathcal{D}_{i,\text{train}} \subseteq \mathcal{D}_{\text{train}}$ is the set of months with observed Y_{it} . Economically, $\hat{\beta}_i$ summarizes firm i 's exposure to latent common components extracted from the characteristic-managed long-short return space.

Out-of-sample prediction. For each $t \in \mathcal{D}_{\text{test}}$, construct $\widetilde{\mathbf{M}}_t^{\text{LS}}$ using the *training* mean $\bar{\mathbf{M}}^{\text{LS}}$ and scaling matrix $\mathbf{D}$, then compute $\hat{\mathbf{f}}_t = \widehat{\mathbf{V}}_K^\top \widetilde{\mathbf{M}}_t^{\text{LS}}$ with fixed $\widehat{\mathbf{V}}_K$. The PCA benchmark forecast is

$$\hat{Y}_{it}^{\text{PCA}} = \hat{\alpha}_i + \hat{\beta}_i^\top \hat{\mathbf{f}}_t. \quad (36)$$

No test-period information is used to re-estimate $\widehat{\mathbf{V}}_K$, $\hat{\alpha}_i$, or $\hat{\beta}_i$, ensuring a strict out-of-sample design.

C.2 Dynamic GNN and integration with NN3 for tangency portfolios

We analyze how a dynamic fitted GNN can contribute to cross-sectional pricing and combine it with a non-graph baseline model to see if their portfolio leads to favorable investment outcomes. In particular, we first construct a model of firm returns using only firm-characteristics information, excluding supply chain graphs and then build a zero-cost portfolio that sorts firms using this model. Our exercise is equivalent to a two-factor Sharpe ratio maximizing portfolio optimization problem with assets $\{F_t, C_t\}$, where F_t is our dynamically fitted GNN-managed portfolio and C_t is the dynamically fitted NN3-managed control risk asset that does not have supply chain graph information.²⁴

The goal is to determine whether the asset F_t provides additional span beyond the control risk factor C_t : if there were any improvement in the efficient portfolio upon adding F_t , it would indicate that the GNN provides pricing information not replicable by the NN3 characteristics-only models.

Expanding window estimation of the GNN. The factor F_t is generated dynamically using an expanding-window training scheme. We begin with an initial sample ending in December 2006. For each subsequent year t , the training set expands to include all data from January 1977 through December of year $t - 1$, i.e.

$$\mathcal{D}_{\text{train},t} = \{(Y_{i\tau}, \mathbf{X}_\tau, SCG_\tau) : \tau \leq t - 1\},$$

and the model is used to forecast returns in calendar year t , using only information available up to each forecast date. In each step, we re-estimate the model parameters Θ_t by minimizing the mean squared error:

$$\min_{\Theta_t} \mathbb{E}_{(i,\tau) \sim \mathcal{D}_{\text{train},t}} \left[(Y_{i\tau} - g_i(\Theta_t; \mathbf{X}_\tau, SCG_\tau))^2 \right].$$

The resulting forecasts induce time-varying factor realizations F_t . This expanding-window design reflects how an investor's conditioning set grows with the arrival of new supply-chain data, while

²⁴ The benchmark C_t is the NN3 long-minus-short portfolio constructed following Equation (23) (§4.1), reported in Table 1. Any model with well-documented out-of-sample performance can be swapped in as the control risk factor.

avoiding the recency bias of fixed windows.

Portfolio construction. At each period, we form the Sharpe ratio maximizing portfolio of $\{F_t, C_t\}$. The Sharpe ratio maximizing weights is $\mathbf{w}^* = \mathbf{\Sigma}^{-1}\boldsymbol{\mu}$. With excess returns, the maximum Sharpe ratio is given compactly by $\text{Sharpe}^* = \sqrt{\boldsymbol{\mu}^\top \mathbf{\Sigma}^{-1} \boldsymbol{\mu}}$, where $\boldsymbol{\mu} = \mathbb{E}[(F_t, C_t)^\top]$ and $\mathbf{\Sigma} = \text{Var}[(F_t, C_t)^\top]$ are estimated in an expanding fashion consistent with the factor construction. This setup ensures that the investor uses the maximum available historical information at each date.

In addition, to incorporate trading costs into our performance evaluation, we also compute adjusted portfolio returns by subtracting proportional transaction costs incurred at each rebalancing. Let $r_{p,t}$ denote the gross return of portfolio p at time t , and let $\Delta w_{p,t} = \|w_{p,t} - w_{p,t-1}\|_1$ denote the turnover implied by portfolio rebalancing between periods $t-1$ and t in absolute values. With a proportional cost rate κ , the net return is given by $\tilde{r}_{p,t} = r_{p,t} - \kappa \cdot \Delta w_{p,t}$. Using these trading cost-adjusted returns, the Sharpe ratio with transaction costs is computed as $SR_p(\kappa) = \frac{\mathbb{E}[\tilde{r}_{p,t}]}{\sqrt{\text{Var}(\tilde{r}_{p,t})}}$. We compute $SR_p(\kappa)$ for $\kappa \in \{1\%, 20\%\}$ to assess robustness to realistic levels of trading costs.

Empirical Results. We compare the out-of-sample performance of three portfolio strategies: the GNN-only factor portfolio ($w_F = 1$), the NN3-managed portfolio ($w_C = 1$), and the Sharpe ratio maximizing portfolio $\mathbf{w}^*$ that combines both sets of factors optimally.

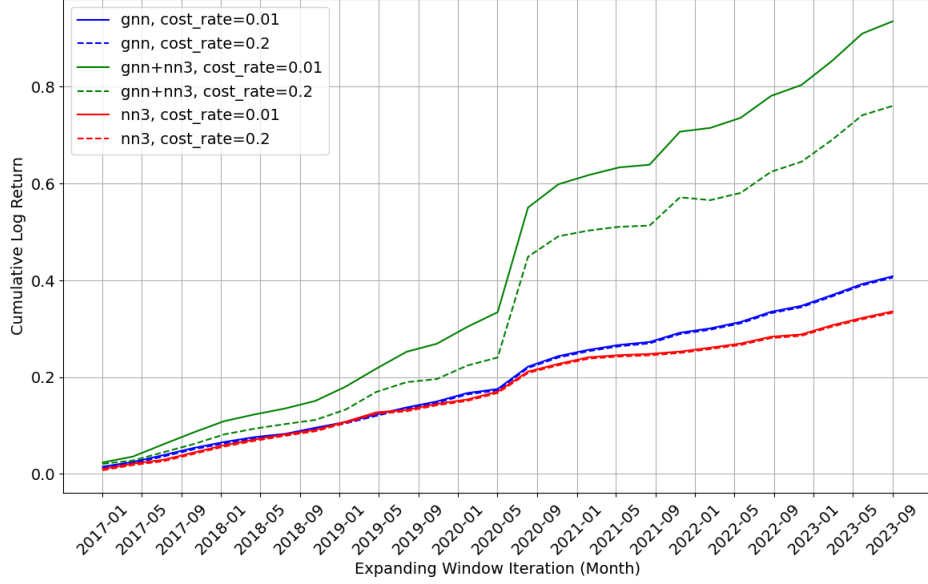
We evaluate the performance of the portfolios using two complementary perspectives: cumulative returns over time and Sharpe ratios that account for trading costs. Together, these results highlight both the incremental value of including the GNN-managed portfolio and the robustness of our findings to transaction cost assumptions.

Figure A.1 plots cumulative log returns under the expanding-window implementation, in which models are re-estimated annually and portfolios rebalanced accordingly. Two patterns are particularly striking. First, the combined GNN+characteristic portfolio (green) consistently dominates both the characteristics-only (red) and GNN-only (blue) portfolios throughout the out-of-sample period, confirming that the GNN factor contributes incremental value beyond the characteristics-only benchmark. Second, the green curves remain clearly above the red curves even at the 20% cost rate, establishing that the GNN’s pricing advantage is not eliminated by transaction cost frictions.

Remark. Our empirical exercise is deliberately parsimonious. Restricting attention to a two-asset frontier avoids overfitting and isolates the marginal contribution of the GNN factor. Moreover, in the presence of estimation error, one may regularize $\mathbf{\Sigma}^{-1}$ via shrinkage or rolling windows; our baseline adopts expanding-window moments with standard stability checks. Replacing NN3 with any alternative characteristics-only construction changes only the definition of C_t ; the Markowitz algebra and inference remain unchanged.

The GNN-managed portfolio is a priced risk factor: adding it to a characteristics-only portfolio expands the mean-variance frontier, with higher-order network aggregation at $d = 3$ as the principal

Figure A.1: Cumulative Log Returns of GNN and NN3 Portfolios, Expanding Window



This figure plots the cumulative log returns of three strategies: the firm characteristic-only portfolio (red), the GNN-only portfolio (blue), and the combined GNN + firm characteristic portfolio (green), under two levels of proportional trading costs (solid: $\kappa = 1\%$; dashed: $\kappa = 20\%$). The figure shows cumulative *log* returns, not raw monthly returns; individual monthly returns fluctuate around zero in all strategies. We use data from January 2017 through January 2024.

driver of that gain. The factor also exhibits the return concentration documented in [Chen et al. \(2024\)](#), whereby top-decile portfolios substantially outperform bottom-decile portfolios in the cross-section. Both the mean-variance expansion and the Sharpe ratio advantage of the GNN-only factor persist under proportional trading costs as high as 20%.

C.3 Correlation with Known Risk Factors

Low correlation with established risk factors is a necessary condition for the supply chain GNN factor to represent a genuinely distinct source of systematic risk. Table [A.1](#) reports pairwise correlations between the GNN-managed SCG portfolio which is constructed using the expanding-window estimates from [§C.2](#), and the three Fama–French factors.

Table A.1: Correlation between SCG factor and Fama–French three factors.

	SCG	Market (MKT)	Size (SMB)	Book-to-Market (HML)
SCG	1			
Market (MKT)	0.179	1		
Size (SMB)	0.187	0.392	1	
Book-to-Market (HML)	0.103	0.157	0.284	1

This table reports the correlations between our GNN-managed portfolio returns generated from expanding window-GNN with the three Fama-French factors (Fama and French, 1993) of Market, Size and Value. GNNs are estimated with $d = 3$. We use data from January 2017 through January 2024, same as the expanding window exercise in previous section.

The SCG factor’s correlations with the three Fama–French factors are low: 0.179 with the Market factor, 0.187 with Size, and 0.103 with Book-to-Market, averaging 0.16 across the three. This average is approximately 56% of the mean pairwise correlation among the Fama–French factors themselves (0.28), which range from 0.16 between Market and Book-to-Market to 0.39 between Market and Size. The Fama–French factors therefore share substantially more common variation with each other than any of them shares with the SCG factor, indicating that the supply chain GNN captures a dimension of return variation that is largely orthogonal to the traditional factor space.

The SCG factor is not a linear recombination of size, value, and market exposures. Rather, it reflects a distinct source of systematic risk arising from network risk transmission that the Fama–French framework does not span, and its low factor correlations imply meaningful diversification benefits when added to a standard factor portfolio.

C.4 Additional Tables and Figures

Additional Tables and Figures for §3

Table A.2: Summary statistics for firms in the manufacturing and wholesale industries

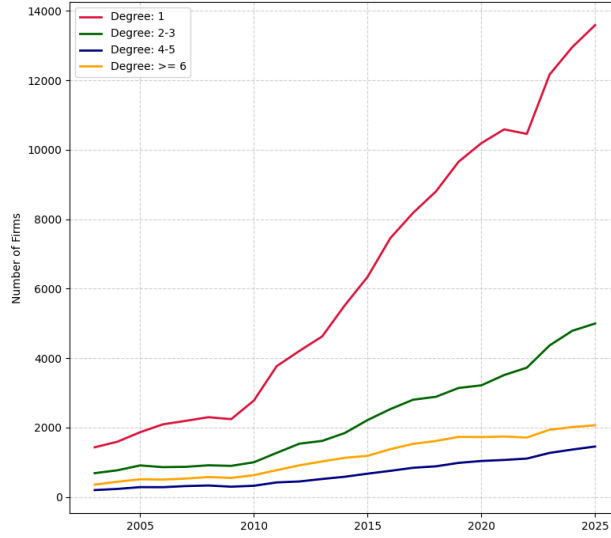
Year	Manufacturing			Wholesale & Retail Trade		
	Num. links	Num. firms	Average degree	Num. links	Num. firms	Average degree
2003	1586	1036	1.53	205	195	1.05
2004	1917	1197	1.60	279	254	1.10
2005	2227	1385	1.61	439	392	1.12
2006	2144	1394	1.54	458	409	1.12
2007	2213	1495	1.48	500	431	1.16
2008	2402	1598	1.50	470	409	1.15
2009	2279	1547	1.47	440	386	1.14
2010	2741	1873	1.46	444	396	1.12
2011	3728	2519	1.48	556	511	1.09
2012	4586	3014	1.52	724	648	1.12
2013	5134	3255	1.58	816	717	1.14
2014	5807	3651	1.59	961	819	1.17
2015	6377	4160	1.53	1273	1076	1.18
2016	7498	4820	1.56	1654	1314	1.26
2017	8569	5414	1.58	1829	1451	1.26
2018	9819	5932	1.66	1930	1563	1.23
2019	10787	6469	1.67	2129	1714	1.24
2020	11185	6856	1.63	2545	2060	1.24
2021	11958	7381	1.62	2671	2173	1.23
2022	12267	7649	1.60	2386	1985	1.20
2023	13804	8659	1.59	2574	2120	1.21
2024	11020	7176	1.54	2181	1830	1.19

We report the number of edges of all nodes and the number of nodes in the corresponding industry, as well as average degree. The number of nodes reflect the number of firms and the number of edges reflect the number of observed supply chain links in our data.

Table A.3: Monthly cross-sectional test R^2 summary statistics for TransformerGNN models.

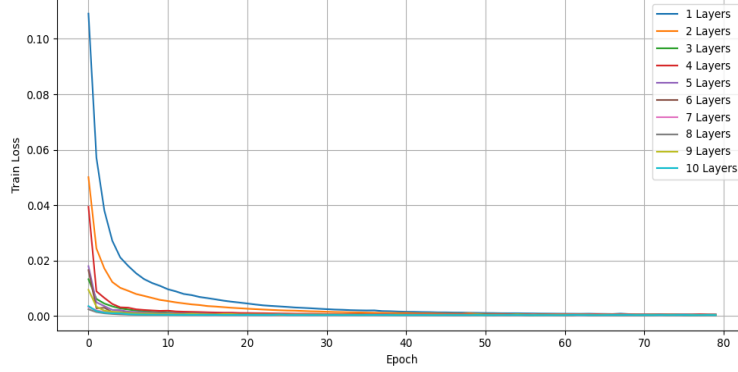
Model	Mean	Std	Min	Max
TransformerGNN ($d = 1$)	0.0034	0.0226	-0.0536	0.0621
TransformerGNN ($d = 2$)	0.0101	0.0170	-0.0566	0.0587
TransformerGNN ($d = 3$)	0.0143	0.0248	-0.1601	0.0970
TransformerGNN ($d = 4$)	0.0133	0.0288	-0.2012	0.1008
TransformerGNN ($d = 5$)	0.0085	0.0370	-0.2887	0.0917
TransformerGNN ($d = 6$)	-0.0221	0.1117	-0.8353	0.0771
TransformerGNN ($d = 7$)	-0.1039	0.2401	-1.6026	0.0314
TransformerGNN ($d = 8$)	-0.1733	0.2891	-1.7702	0.0004
TransformerGNN ($d = 9$)	-0.1734	0.2908	-1.7777	0.0005
TransformerGNN ($d = 10$)	-0.1734	0.2881	-1.7650	0.0006
NN (2 hidden layers)	-0.0067	0.1571	-0.8021	0.1969
NN (3 hidden layers)	-0.0073	0.0953	-0.5366	0.1422
NN (10 hidden layers)	-0.1854	0.3438	-1.9918	-0.0000

This table reports out-of-sample predictive R^2 for our GNN models and various benchmark models. We report the results above for the out-of-sample period of January 2017 through January 2024 to avoid look-ahead bias. We present TransformerGNN models with different neighborhood radii d , where d controls the number of message-passing layers, in addition to standard multi-layer perceptron neural network (NN) benchmarks without graph structure.

Figure A.2: Number of firms in each group of degrees across time.

The x-axis represents the time, while the y-axis shows the number of firms. Each line corresponds to a different group. This figure reveals the complex temporal dynamics in the number of firms within each group. To address this, we sample an equal number of nodes from each group for our Monte Carlo experiments, ensuring that our analysis is not biased toward the group with the largest node count.

Figure A.3: Training loss over batches for different model configurations.



The x-axis represents the number of batches, while the y-axis shows the training loss. We use data in the period ranging from January 2003 to December 2016. Each line corresponds to a different model variant. From this figure, we can observe that trainings have converged for all the models.

Additional Tables and Figures for §4

Table A.4: Out-of-Sample Hansen–Jagannathan Distance: JKP 13 Anomaly Theme Factors

Model	$d_{HJ,OS}$	HJD_{OS}	Reduction vs. NN3
Characteristics-only NN3	0.3410	0.1163	—
Supply chain GNN only	0.3439	0.1182	-1.63%
GNN + NN3 (combined)	0.3363	0.1131	2.75%

Out-of-sample Hansen–Jagannathan distance (HJD) for models evaluated against the 13 anomaly theme factors of [Jensen et al. \(2023\)](#) (JKP13). The HJD measures the joint magnitude of pricing errors across all test portfolios; lower values indicate better pricing. We use the data period from January 2017 through January 2024. $d_{HJ,OS}$ is the HJ distance and $HJD_{OS} = d_{HJ,OS}^2$. “Reduction vs. NN3” is the percentage improvement relative to the characteristics-only multilayer perceptron baseline; negative values indicate underperformance relative to that baseline. Point estimates are reported without formal standard errors; given the short out-of-sample window, modest reductions (e.g., -1.63% for the GNN-only model) should be interpreted with caution.

Table A.5: Out-of-Sample Hansen–Jagannathan Distance: Fama–French 25 Size–Value Portfolios

Model	$d_{HJ,OS}$	HJD_{OS}	Reduction vs. NN3
Characteristics-only NN3	0.3235	0.1046	—
Supply chain GNN only	0.3189	0.1017	2.77%
GNN + NN3 (combined)	0.3183	0.1013	3.15%

Out-of-sample HJD for models evaluated against the 25 Fama–French portfolios double-sorted on size (market equity) and book-to-market ratio ([Fama and French, 1993](#)). We use the data period from January 2017 through January 2024. See Table A.4 for column definitions.

Table A.6: Out-of-Sample Hansen–Jagannathan Distance: Fama–French 30 Industry Portfolios

Model	$d_{HJ,OS}$	HJD _{OS}	Reduction vs. NN3
Characteristics-only NN3	0.3711	0.1377	–
Supply chain GNN only	0.3379	0.1142	17.07%
GNN + NN3 (combined)	0.3374	0.1138	17.36%

Out-of-sample HJD for models evaluated against the 30 Fama–French industry portfolios (Fama and French, 1997). See Table A.4 for column definitions. We use the data period from January 2017 through January 2024. The large HJD reduction for the supply chain GNN on industry portfolios (17%) reflects the close alignment between industry-level risk exposure and supply chain network structure.

Table A.7: Net return (%) by cost rate κ and holding horizon h .

$k \setminus h$	$h = 0$	$h = 1$	$h = 3$	$h = 6$
0.0000	1.139	0.773	0.562	0.734
0.0001	1.106	0.740	0.552	0.728
0.0003	1.041	0.673	0.531	0.717
0.0010	0.811	0.439	0.457	0.678
0.0030	0.156	-0.228	0.247	0.568
0.0100	-2.138	-2.563	-0.489	0.181

This table reports net monthly-equivalent returns (%) for the decile long-minus-short strategy under the (κ, h) block-rebalancing experiment of Jensen et al. (2024). We use the data period from January 2017 through January 2024. Rows are proportional transaction cost rates κ (expressed as fractions; e.g., 0.0100 = 1%). Columns are holding horizons $h \in \{0, 1, 3, 6\}$ months, where $h = 0$ corresponds to monthly rebalancing. Each cell is computed from the test sample as the average monthly-equivalent net return. For a block of length h , the net return deducts κ times the turnover incurred at the rebalance. Negative values indicate that transaction-cost frictions to return exceeds gross return at that (κ, h) combination.

Table A.8: Sharpe ratio by cost rate κ and holding horizon h .

$k \setminus h$	$h = 0$	$h = 1$	$h = 3$	$h = 6$
0.0000	0.156	0.116	0.155	0.241
0.0001	0.152	0.111	0.152	0.239
0.0003	0.143	0.101	0.147	0.236
0.0010	0.111	0.066	0.126	0.223
0.0030	0.021	-0.034	0.068	0.187
0.0100	-0.278	-0.366	-0.136	0.060

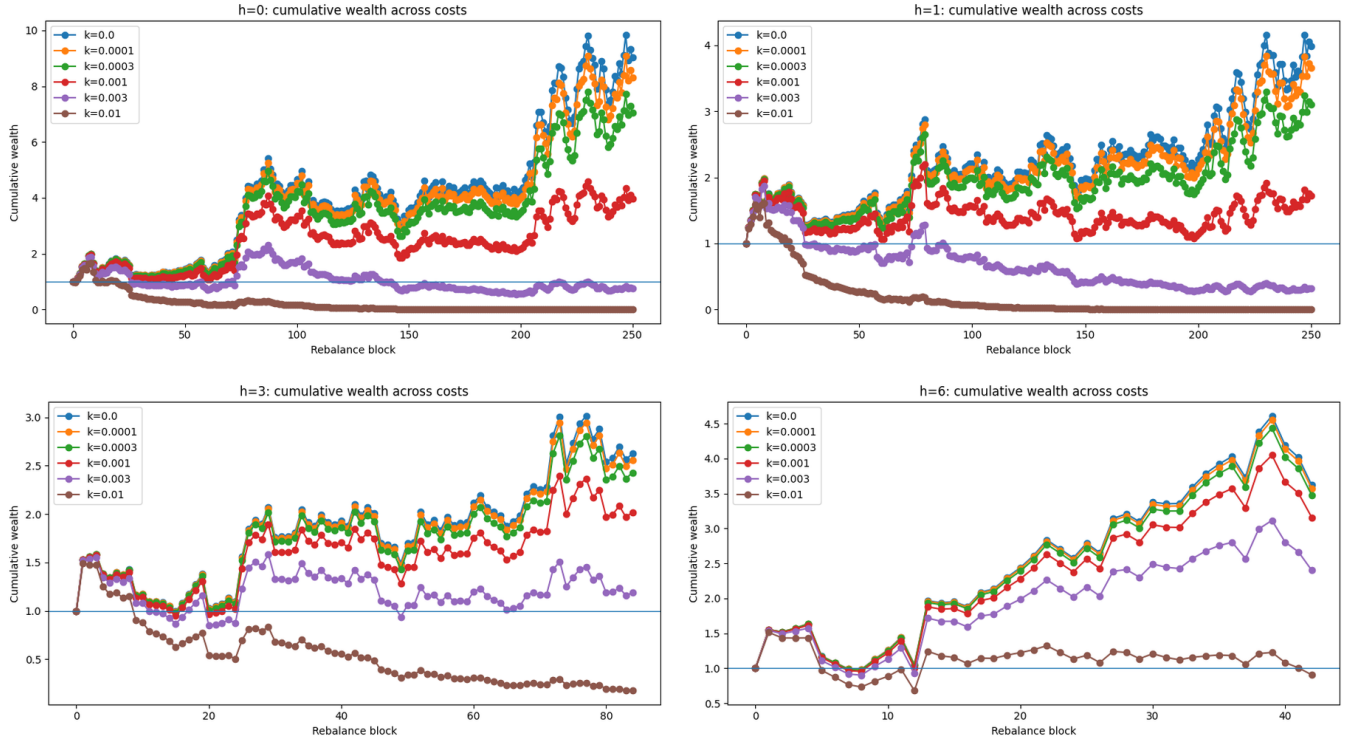
This table reports monthly-equivalent Sharpe ratios computed on net-of-cost returns under the (κ, h) block-rebalancing experiment. We use the data period from January 2017 through January 2024. Rows are proportional transaction cost rates κ ; columns are holding horizons h , following the same convention as Table A.7. For each (κ, h) cell, the Sharpe ratio is computed from the test sample as the mean monthly-equivalent net return divided by the standard deviation of monthly-equivalent net returns. Sharpe ratios differ from those in Table 6 because Table 6 uses the standard equal-weighted decile construction of Equation (23); the differences reflect portfolio-construction protocol rather than signal content.

Table A.9: Average turnover per rebalance block by holding horizon h .

$h = 0$	$h = 1$	$h = 3$	$h = 6$
3.277	3.336	3.154	3.316

This table reports average portfolio turnover per rebalance block for each holding horizon h . We use the data period from January 2017 through January 2024. Turnover is invariant to κ because portfolio construction (decile sorting on predicted returns) does not depend on the cost rate; hence all rows within a column are identical. The effective annualized turnover cost for a given (κ, h) pair is approximately $\kappa \times \text{turnover} \times (12/h)$, which falls as h increases even though per-block turnover is similar across horizons.

Figure A.4: Cumulative Returns under Multi-Step Forecasting Horizons



This figure plots cumulative log returns of the decile long-minus-short strategy over the out-of-sample period $\mathcal{D}_{\text{test}}$ for each holding horizon $h \in \{0, 1, 3, 6\}$ months. Each panel corresponds to a proportional transaction cost rate κ . The $h = 0$ strategy dominates at low cost rates, while the $h = 6$ strategy dominates at high cost rates, consistent with the crossover visible in Figure 6a. We use the data period from January 2003 through January 2024.

Additional Tables for §5

Table A.10: Mean Average Distance (MAD) of Final Layer Embeddings by Network Depth

Network Depth d	$MAD^{(d)}$	Median
1	2.6110	2.4210
2	1.3727	1.2897
3	0.1202	0.1101
4	0.1103	0.1028
5	0.0270	0.0237
6	0.0312	0.0294
7	0.0137	0.0134
8	0.0061	0.0061
9	0.0179	0.0179
10	0.0105	0.0105

This table reports the average and median of the cross-sectional monthly Mean Average Distances (MAD) of final layer embeddings for TransformerGNN models with different depths d . MAD is computed as the average pairwise Euclidean distance between all firm embeddings at the last layer, as defined in Equation (24). Lower MAD values indicate greater smoothing, with values approaching zero indicating complete loss of firm-specific heterogeneity. We compute the test embeddings which cover the out-of-sample period ranging from January 2017 through January 2024.